

Shipment Networks and Market Power in U.S. Poultry Procurement

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Abstract

How do market power and geography shape the cost of supply-chain shocks in U.S. poultry procurement? Linking Commodity Flow Survey records to Census establishment microdata, I estimate random-coefficients nested logit demand with multi-product Bertrand supply. Wholesale buyers are roughly three times as price sensitive as food-service buyers, and shipment distance raises marginal cost while lowering differentiated-product demand. Counterfactually, an industry-wide raw-poultry cost shock lowers total surplus by 17.7 percent and localized plant shutdowns by 4.8 percent, with losses concentrated in exposed destinations; a merger of the third- and fourth-largest firms converts 59 percent of downstream buyers' losses into deadweight loss. The estimates provide, to my knowledge, the first structural demand-and-supply benchmarks for the processor-to-buyer layer of a major U.S. meat industry—the layer on which recent merger enforcement and information-exchange litigation in poultry operate.

JEL Codes: C45, C51, L13, L66, Q13, R41.

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1 Introduction

U.S. poultry markets have drawn sustained attention from competition authorities and supply-chain planners. In 2022, Cargill and Continental Grain acquired Sanderson Farms, then the third-largest U.S. broiler processor, and combined it with Wayne Farms to form a new third-largest firm (Cargill, 2022). The transaction closed under a Department of Justice consent decree, and the same layer of the industry remains under active federal scrutiny: the Department of Justice’s suit against the information firm Agri Stats turns on how processors price shipments to wholesale, retail, and food-service buyers (U.S. Department of Justice, 2023), and private antitrust litigation alleges coordinated supply restrictions over much of the sample period (In re Broiler Chicken Antitrust Litigation, 2016). Tornadoes and hurricanes have repeatedly idled processing plants across the Southeast, and disease outbreaks periodically remove capacity with little warning. Evaluating either consolidation or disruption requires quantitative answers to the same three questions: how easily do downstream buyers substitute across plants and products, how much pricing power do processors hold over those buyers, and who bears the losses when part of the supply network fails? Yet the layer of the supply chain where these questions live—shipments from processing plants to wholesalers, retailers, and food-service buyers—is nearly invisible in standard data sources.

This paper opens that layer using restricted-use Census microdata and asks how market power and geography jointly determine the welfare cost of supply-chain shocks in U.S. poultry procurement. The setting is downstream procurement rather than household retail choice: buyers in a destination county source poultry from competing origin establishments and product categories. Products are establishment–commodity pairs, and markets are destination-county-year cells. This adapts the differentiated-products framework of Berry (1994) and Berry, Levinsohn, and Pakes (1995) to freight microdata, where differentiation comes from plants, geography, shipment attributes, and commodity class.

The empirical foundation links Commodity Flow Survey (CFS) shipment records to Census establishment microdata. I use CFS records to construct a poultry shipment network, classify ambiguous processor-origin shipments into processor-to-retailer and processor-to-wholesaler links with a random forest trained on channel labels derived from Longitudinal Business Database (LBD) destination composition, and assign wholesaler-origin shipments to downstream-distribution links by construction. The resulting network measure, ToW, enters demand as a wholesale-channel share. I then use LBD food-service, retail, and wholesale establishments to build market-specific representative buyer records, so that the demand system can reflect each market’s downstream buyer mix. The structural model estimates random-coefficients BLP demand with product-category nests and buyer-type-specific price

sensitivity, recovers marginal costs under multi-product Bertrand pricing, and simulates plant-unavailability, distance, input-cost, and ownership shocks.

Three findings characterize baseline competition. First, procurement demand is elastic—the average own-price elasticity is about -3.1 —and shipment distance matters twice: it raises recovered marginal cost for all products, and it lowers mean utility for differentiated products, while for commodity broilers the demand-side penalty nets out and geography operates through the cost channel alone. Second, buyer composition matters. The estimates attribute to retail buyers about twice the price sensitivity of food-service buyers, and to wholesale buyers about 3.3 times as much; because receiver-level choices are not observed, this is model-based heterogeneity, identified from the covariation of market demand with each market’s calibrated buyer mix. This ranking is consistent with the position each buyer type occupies in the supply chain: merchant wholesalers resell near-commodity product on thin operating margins and monitor quotations across many competing suppliers, so procurement prices map almost one-for-one into their resale margins, while food-service demand is buffered by menus and product specifications. Third, recovered price–cost margins average roughly 33 percent of the unit value, close to the level implied by the demand elasticity alone; multi-product ownership adds a modest increment on average but concentrates pricing power where portfolios overlap within markets.

The counterfactuals quantify four distinct adjustment channels. Removing the products of a weather-exposed set of processing counties lowers total surplus by 4.8 percent, and a severe logistics shock that scales log distance by 1.2 lowers it by 4.6 percent; in both cases buyers partially reallocate toward unaffected plants, and losses concentrate in the most exposed destinations. An industry-wide 20 percent raw-poultry cost increase, which reaches every supplier, lowers total surplus by 17.7 percent. Because the four shocks differ in primitive scale, these figures rank policy-relevant scenarios and contrast mechanisms; they do not measure the intrinsic value of spatial substitution. A merger of the third- and fourth-largest firms leaves cost primitives unchanged but raises average unit values by about 7 percent; of the 6.53 billion dollar loss to downstream buyers, 2.70 billion is transferred to producers and 3.83 billion—59 percent—is deadweight loss. This scenario approximates the 2022 Sanderson–Wayne consolidation that opens the paper, which the Department of Justice addressed with a consent decree rather than a merger challenge ([U.S. Department of Justice, 2022](#)); the decomposition therefore speaks directly to the class of transactions that merger review in this industry actually confronts.

Two boundaries define what the exercise measures. First, the market power estimated here is seller-side: the pricing power of multi-plant processing firms over their downstream buyers. It is the complement of the buyer power studied on the other side of the processing

bottleneck. Recent structural work on beef packing estimates oligopsony markdowns in fed-cattle procurement while treating the downstream wholesale price as externally determined (Garrido et al., 2024; Moschini and Smith, 2026), and the agricultural-economics literature on production contracts examines the integrator–grower relationship (Vukina, 2001; MacDonald, 2014). This paper supplies the sales side that those studies hold fixed—and in poultry that side is the natural object of study: 97 percent of broilers are grown under contract and no live-bird spot market exists—live birds move to plants through internal transfer pricing (Weng, Vukina, and Zheng, 2015)—so the upstream price interface modeled in beef is largely internalized within the integrator, and the industry’s competitive interface sits downstream at the plant gate. The upstream grow-out relationship enters here only as motivation for the plant-unavailability scenario. Relatedly, the static Bertrand–Nash assumption on FOB-plant unit values is a benchmark conduct model. Subsection 2.2 describes the pricing institutions behind it: much poultry commerce runs on formula contracts indexed to wholesale quotations or on recurring negotiated agreements, so competitive conditions transmit into transaction prices at the model’s annual frequency, while quantity commitments and bilateral bargaining with large buyers remain outside the model; recovered margins summarize average pricing incentives rather than transaction-level conduct. Second, the CFS records shipments to destination locations, not household purchases. The demand side is downstream procurement demand, and buyer surplus below is downstream-buyer surplus, not household consumer surplus.

The paper contributes to three strands of work. To the empirical industrial organization of differentiated products, it shows that freight microdata can support a complete demand-and-supply system: relative to spatial pricing studies built on plant-level shipments (Miller and Osborne, 2014) and to the merger-simulation tradition (Nevo, 2000b; Miller and Weinberg, 2017), the contribution is a procurement-level market definition with observed plant costs and buyer-composition heterogeneity. The closest structural analog, Wollmann (2018), combines buyer-type demand heterogeneity with multi-product Bertrand supply for commercial trucks, estimated from vehicle registrations and list prices; the present system is estimated on observed shipment-level transaction values, and, to my knowledge, it is the first complete demand-and-supply system for the processor-to-buyer layer of a major U.S. meat industry. The closest data antecedent, Ganapati (2025), combines similar Census freight and establishment linkages to explain the rise of large wholesalers across manufacturing; here the same infrastructure instead identifies how a food manufacturer prices to heterogeneous downstream buyers. To the agricultural-economics literature on imperfect competition in food markets (Sexton, 2013; Çakır and Balagtas, 2012), on vertical relationships in the food chain (Villas-Boas, 2007; Bonnet and Dubois, 2010), and to its recent

structural turn—oligopsony models of cattle procurement (Garrido et al., 2024; Moschini and Smith, 2026) and production-function decompositions of meatpacking margins (Azzam, Jaumandreu, and Lopez, 2025)—it adds the demand system those approaches lack, at the plant-to-buyer layer where merger review, the Packers and Stockyards rulemaking agenda, and the federal information-exchange litigation actually operate (U.S. Department of Agriculture, Agricultural Marketing Service, 2024; U.S. Department of Justice, 2023). And to work on supply-chain disruption, it complements reduced-form evidence on shock propagation with an equilibrium framework in which the same network that transmits shocks also disciplines market power. Machine learning enters as a data-construction tool for unobserved shipment channels, in the sense of Mullainathan and Spiess (2017).

The rest of the paper proceeds as follows. Sections 2 and 3 describe the market setting, data, network classification, and representative-buyer construction. Section 4 presents the model and identification strategy. Sections 5 and 6 report estimates and counterfactuals. Section 7 discusses limitations and concludes. Appendix A collects the tables and figures. Appendices B and C report data-construction details and variable definitions, Appendix D reports state-level counterfactual results, and Appendix E reports disclosure-constrained validation checks.

2 Institutional Background and Market Definition

2.1 Poultry Shipment Channels

Poultry products move through a spatial supply chain that links processing plants, wholesalers, retailers, and food-service buyers. I use the notation P for poultry processors, W for wholesalers, R for retailers, and F for food-service establishments. The descriptive network focuses on three directed shipment channels: processor-to-retailer (PR) links, processor-to-wholesaler (PW) links, and wholesaler-to-retailer or downstream-distribution (WR) links, the last covering shipments from wholesalers to retail or food-service destinations. Distinguishing these links matters because wholesale and retail channels differ in shipment size, distance, inventory behavior, and likely substitution opportunities. A shipment from a processor to a wholesaler is not economically equivalent to a shipment from the same processor directly to a retailer, even if the commodity code is the same.

Poultry products are also differentiated by commodity form, processing technology, temperature-control requirements, origin plant, and destination. Fresh and chilled poultry shipments differ from frozen shipments because cold-chain constraints, delivery timing, and inventory management differ across these forms. Distance can therefore affect demand

through delivery reliability and product quality as well as through marginal cost. This emphasis on delivery timing is consistent with trade evidence that time in transit can itself operate as a trade barrier (Hummels and Schaur, 2013). In a freight dataset, these demand and cost channels are observed through FOB-plant shipment unit values, weights, and distances rather than through scanner-level retail choices.

2.2 Pricing Institutions in Poultry Procurement

How poultry is priced matters for how competition should be modeled. Much of U.S. poultry moves under supply agreements rather than one-off transactions, and the available institutional evidence indicates that these agreements often fix quantities, specifications, and delivery terms while letting prices adjust: formula contracts indexed to published wholesale quotations are widespread, and prices outside such formulas are typically re-set through recurring, often annual, bids and negotiations. Retail chill-pack programs at major processors have been sold under multi-year agreements with formula-linked prices (MEAT+POULTRY, 2016); products destined for food service are commonly priced off daily wholesale quotations or awarded through periodic bids, and sales to distributors follow similar arrangements. Because poultry is not covered by mandatory livestock price reporting, privately reported quotations—most prominently the daily Urner Barry poultry quotations—serve as the industry’s de facto price benchmark (Urner Barry, n.d.; Adjemian et al., 2016). This is a substantive contrast with beef, where mandatory livestock price reporting underpins both the public price series and the recent structural literature on packer conduct (Garrido et al., 2024). Poultry is thus a test case for measuring market power in a major meat industry without mandatory reporting, and the restricted-use shipment microdata used here substitute for public series that do not exist. The transactions behind those quotations come from a comparatively thin spot layer: merchant wholesalers, traders, and cold-storage holders who buy surplus loads and commodity items and can redirect purchases across suppliers at short notice.

These institutions motivate the paper’s conduct assumption. Because contract prices are widely indexed to current quotations or re-set through recurring negotiations, competitive conditions transmit into transaction prices within the model’s annual time unit—the price flexibility that static Bertrand–Nash competition in FOB-plant unit values is designed to capture. The FOB-plant unit value used in estimation aggregates a product’s transactions within a market and survey year, so it averages over contract and spot execution rather than isolating either mechanism. The institutions also identify which buyers arbitrage prices in the short run: wholesalers and traders monitor quotations across many suppliers, whereas

retail and food-service buyers are committed to programs and product specifications within a season. The demand estimates in Section 5 are consistent with this margin, since wholesale-type buyers are estimated to be by far the most price-sensitive buyer type.

The frequency of the model matters for how the conduct assumption should be read. The model’s time unit is the survey year: the price entering the pricing first-order conditions is a product’s CFS-expanded average transaction price over the year, not a posted weekly quote. At that frequency, formula indexation, periodic renegotiation of discounts, and reallocation of volume across customers give firms effective control over their average price levels, so static Nash pricing at the frequency of the data is the natural benchmark—the same interpretation under which posted-price competition is routinely applied to averaged transaction prices in other differentiated-products settings (Nevo, 2001; Miller and Osborne, 2014). Retail evidence from another heavily benchmarked food-and-beverage category points the same way: distilled-spirits prices are rigid and adjust in discrete increments at weekly frequency, yet the implied frictions resolve within the year (Conlon and Rao, 2020), so annual averaging integrates over short-run pricing rigidities that would complicate a higher-frequency conduct model. Poultry procurement is also a repeated flow purchase with limited storability outside frozen product, so the dynamic considerations that complicate static pricing in durable- or storable-goods markets are muted.

The two leading deviations from the unilateral benchmark move the recovered objects in opposite directions, so their biases work against each other rather than compounding. Bilateral bargaining by the largest retail and food-service buyers would place transaction prices below sellers’ unilateral best responses; the first-order conditions would then attribute those discounts to lower marginal cost, and the Bertrand margins would overstate the surplus sellers actually retain from such buyers. Conversely, the sample period overlaps the conduct period alleged in private and federal antitrust litigation concerning coordinated supply restrictions among broiler producers (*In re Broiler Chicken Antitrust Litigation*, 2016); to the extent conduct was more accommodating than unilateral pricing, the first-order conditions attribute the coordinated component of prices to marginal cost, and the reported margins understate market power. The same direction of bias applies to the information-exchange conduct alleged in the Department of Justice’s suit against Agri Stats, which concerns processors’ benchmarking of prices charged to downstream buyers (U.S. Department of Justice, 2023). The unilateral benchmark therefore carries evidentiary value of its own: the recovered margins measure the pricing incentives that annual-frequency competition would generate absent coordination—a structural reference point that the litigation record otherwise lacks, and a conservative lower bound on market power during the alleged conduct period. Estimating either alternative directly would require downstream resale prices, contract terms,

or bilateral transaction data that the CFS does not provide ([Villas-Boas, 2007](#); [Bonnet and Dubois, 2010](#)). I therefore hold the unilateral benchmark fixed and read the recovered margins as summarizing average pricing incentives rather than transaction-level conduct.

2.3 Structural Procurement Markets

The structural model defines products as establishment–commodity pairs and markets as destination-county-year procurement markets. This market definition differs from the descriptive ZIP-code network. The ZIP-code network is used to document the upstream shipment structure, classify PR and PW channels, and construct network-based variables such as ToW. The BLP demand and supply model then aggregates to destination counties because the structural choice set must contain multiple competing processor-origin products serving the same downstream procurement market.

This distinction is important for interpretation. The descriptive network asks how poultry shipments are organized across ZIP-code links and downstream channels. The structural model asks how destination-county buyers substitute among competing establishment–commodity products and how firms price those products under common ownership. The two pieces are therefore complementary: the network provides channel and geography information, while the BLP model turns that information into demand, markup, and welfare objects.

The poultry industry is organizationally distinctive. Broiler production has long relied on vertical coordination between growers, integrators, and processing establishments, and processing firms often operate multiple establishments ([Martinez, 1999](#); [Vukina, 2001](#); [MacDonald, 2008](#)). Concentration has also risen steadily: in Census of Manufactures data, the four-firm concentration ratio of broiler processing climbed from 44 to 59 percent between 1992 and 2007 ([Weng, Vukina, and Zheng, 2015](#)), and the 2022 Sanderson–Wayne transaction subsequently combined the industry’s third-largest producer with a mid-sized rival ([U.S. Department of Justice, 2022](#)). Poultry nevertheless remains the least concentrated of the major U.S. meat industries—four-firm ratios in beef packing exceed 80 percent—which makes it a setting where market power must be measured rather than presumed from structure alone. More broadly, modern agricultural markets often combine product differentiation, vertical coordination, and concentration, so market power cannot be evaluated from atomistic textbook assumptions alone ([Sexton, 2013](#)). These features imply that pricing incentives may depend on both plant-level cost conditions and firm-level ownership links. A multi-plant firm selling more than one poultry product into the same destination market internalizes some substitution across its own products when setting prices, while rival-owned plants do not.

The upstream grow-out side is also not a spot market for live birds. Broiler production is organized through production contracts in which integrators supply chicks, feed, veterinary services, technical assistance, catching, and live-haul services, while growers provide housing capital, labor, utilities, and day-to-day husbandry ([MacDonald, 2014](#); [Vukina, 2001](#)). USDA survey evidence reports that 97 percent of broilers were raised on contract operations in 2011. The same evidence shows that grower–plant geography is highly local: the mean distance from a grower to the integrator’s processing plant was 34 miles, and 90 percent of birds were produced within 60 miles of the plant ([MacDonald, 2014](#)). Because growers make long-lived, specialized investments in poultry houses and usually face only a few nearby integrators, upstream supply is better viewed as a governed contractual relationship than as an unconstrained spot market. This relationship is itself an active regulatory margin: recent Packers and Stockyards Act rulemaking targets tournament-payment disclosure and grower compensation ([U.S. Department of Agriculture, Agricultural Marketing Service, 2023](#)), and the Sanderson–Wayne transaction closed under a consent decree addressing grower pay ([U.S. Department of Justice, 2022](#)). Those instruments regulate the upstream boundary of the markets studied here; the downstream sales side, by the USDA’s own account, has seen decades of little enforcement attention ([U.S. Department of Agriculture, Agricultural Marketing Service, 2024](#)). This institutional structure motivates the focus on weather-exposed poultry-processing and production counties in the plant-unavailability counterfactual: a local disruption can interrupt nearby grower relationships and live-bird flows as well as downstream shipments, although the simulation in Section 6 models only the downstream choice-set consequences.

2.4 Related Literature

The modeling approach follows the random-coefficients logit tradition introduced by [Berry \(1994\)](#) and [Berry, Levinsohn, and Pakes \(1995\)](#) and used in supply-side markup estimation by [Nevo \(2001\)](#). The paper also follows practical estimation guidance in [Nevo \(2000a\)](#), [Dubé, Fox, and Su \(2012\)](#), and [Conlon and Gortmaker \(2020\)](#). Identification with market-level data follows the logic in [Berry and Haile \(2014\)](#); the use of local representative-buyer information is related to work on incorporating micro data into differentiated-products demand estimation ([Berry and Haile, 2024](#); [Conlon and Gortmaker, 2025](#)). Here, the representative records describe local establishment composition rather than individual purchase histories. The instrument strategy uses cost shifters and product-crowding measures, including local differentiation instruments in the spirit of [Gandhi and Houde \(2019\)](#).

The shipment-network component connects the paper to work on production networks

and firm-to-firm linkages (Acemoglu et al., 2012; Atalay et al., 2011; Bernard, Moxnes, and Saito, 2019). The ownership dimension is related to evidence on vertical integration and input flows (Atalay, Hortaçsu, and Syverson, 2014). Relative to most differentiated-products applications, the unit of observation here is a business-to-business shipment product in a destination-county procurement market rather than a consumer retail product. Relative to descriptive supply-chain work, the paper uses shipment-network information as an input into a structural demand-and-supply model.

A rapidly growing structural literature examines competition in meat supply chains, concentrated on the procurement side. Garrido et al. (2024) model how formula contracts amplify beef packers’ oligopsony power over fed cattle; Moschini and Smith (2026) estimate spatial buyer power in the same industry; and Azzam, Jaumandreu, and Lopez (2025), using Census plant microdata for meatpacking, decompose gross margins into technology, labor-market, and product-market components, finding no livestock-side buyer power and recovering product-market power only as a scalar residual. All three treat the downstream sales side as exogenous or aggregate. This paper opens that side with a full demand system and buyer-type heterogeneity, in the meat industry where downstream pricing is currently the focus of federal enforcement. In method, Wollmann (2018) and Moschini and Perry (2026) provide the closest precedents—buyer-type demand heterogeneity with multi-product Bertrand supply in commercial trucks, and differentiated-products Bertrand pricing recovered from micro-level purchase data in seed markets—and Ganapati (2025) the closest data antecedent, studying wholesalers themselves rather than manufacturers’ pricing to buyers.

The counterfactuals are also related to evidence that localized shocks can propagate through production networks (Barrot and Sauvagnat, 2016; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021) and to recent theory on how production-network structure shapes disruption losses (Elliott and Jackson, 2025). The empirical contribution here is to combine that supply-chain motivation with a demand-and-supply model that can recompute prices, quantities, markups, and downstream-buyer surplus after plant-unavailability, distance, input-cost, and ownership shocks.

3 Data

This section describes the linked shipment, establishment, and manufacturing data used to construct the estimation sample. I first define CFS-expanded shipment weights and values. I then describe how processor-origin shipments are assigned to downstream channels, and how these assignments provide the wholesale-channel share used later in estimation. The section then defines products, markets, FOB-plant unit values, quantities, shipment

distances, shares, market size, and downstream buyer-composition measures. Appendix B reports additional construction details.

3.1 CFS Shipments and Census Microdata

The Commodity Flow Survey (CFS) is a shipper-based survey conducted jointly by the U.S. Census Bureau and the Bureau of Transportation Statistics. The survey is conducted every five years and provides repeated cross-sectional information on freight shipments originating from selected business establishments. The sampling frame covers establishments in manufacturing, mining, wholesale trade, and selected retail and service industries, together with auxiliary establishments such as warehouses and distribution centers, while excluding several important categories such as farm-origin shipments and imports. Sampled establishments report outbound shipments for assigned one-week periods in each calendar quarter of the survey year, allowing the CFS to capture seasonal variation in freight flows. For each reported shipment, the CFS records commodity type, shipment value, weight, transportation mode, origin, destination, distance shipped, and related shipment characteristics (U.S. Census Bureau, 2025, 2017).

I focus on poultry-related commodities and conduct the analyses using the 2007, 2012, and 2017 waves of the CFS. Poultry shipments are identified using the Standard Classification of Transported Goods (SCTG) commodity codes for fresh or chilled poultry and frozen poultry; live-poultry shipments do not enter the descriptive network or the estimation sample. I use these repeated cross sections to construct poultry shipment networks and compare the geographic structure of poultry flows across survey years.

The CFS reports two distinct weight concepts: the physical shipment weight reported for shipment record s , w_s^{ship} , and the CFS sampling weight that expands the sampled record to the population, κ_s^{CFS} . I use “CFS-expanded shipment weight” to mean their product,

$$\tilde{w}_s = \kappa_s^{\text{CFS}} w_s^{\text{ship}}.$$

Similarly, v_s is the reported shipment value, and $\tilde{v}_s = \kappa_s^{\text{CFS}} v_s$ is the CFS-expanded shipment value. These expanded weights and values are the building blocks for product-market quantities, FOB-plant unit values, weighted distances, and channel shares below.

I link the CFS shipment records to establishment-level businesses through the Longitudinal Business Database (LBD). The LBD is an annual, longitudinal establishment database constructed from the Census Bureau’s Business Register and related administrative records. It covers private, nonfarm employer establishments in the United States and provides longitudinal establishment identifiers, firm identifiers, establishment location, employment, payroll,

and industry classification using the North American Industry Classification System (NAICS) (Jarmin and Miranda, 2002). These NAICS codes help identify processors, wholesalers, retailers, and other downstream establishment types.

I use the longitudinal establishment identifiers to follow plants over time and connect CFS shipment records to plant-level characteristics. Firm identifiers group establishments under common ownership and distinguish single-plant firms from multi-establishment firms. For manufacturing establishments, I further merge plant-level information from the Census of Manufactures (CMF) and the Annual Survey of Manufactures (ASM), which provide measures of employment, payroll, inventories, assets, capital expenditures, materials costs, fuel and electricity expenditures, and detailed product-line shipments. Together, these linked datasets characterize poultry shipments by plant scale, input use, cost shifters, product specialization, and firm ownership structure. They also identify when multiple products or shipments in the same destination market are supplied by establishments under common ownership, which is central for modeling multi-product pricing incentives.

On the demand side, I use LBD establishments in destination counties to describe the local downstream business environment, including food-service, retail, and wholesale establishments. Later subsections use these records to measure downstream buyer composition and to help distinguish retail-oriented from wholesale-oriented shipment destinations. I also use county-level wholesale and retail activity from the Annual Wholesale Trade Survey (AWTS) and Annual Retail Trade Survey (ARTS) as auxiliary information on local downstream market capacity (U.S. Census Bureau, 2024a,b). These linked shipment, establishment, and firm records support a demand-and-supply framework with endogenous FOB-plant unit values, plant-level marginal-cost heterogeneity, multi-product ownership, and market-specific downstream buyer composition.

3.2 Classifying Downstream Shipment Channels

Processor-origin shipments are divided into processor-to-retailer links (PR) and processor-to-wholesaler links (PW). This distinction matters because PR shipments represent direct service from processors to retail destinations, while PW shipments route product through wholesale intermediaries before reaching downstream markets. The main classification challenge arises because the CFS identifies the destination ZIP code of each shipment but does not report the receiver’s establishment identifier. If a destination ZIP code contains both retail and wholesale establishments in the LBD, the shipment cannot be assigned mechanically to either the PR or PW channel.

The PR–PW channel can be assigned mechanically only when the destination ZIP code

contains a single eligible downstream establishment type in the LBD. I use these unambiguous cases to construct a labeled sample for classifying ambiguous processor-origin shipments. Specifically, a processor-origin shipment is labeled PW if the destination ZIP contains eligible wholesalers but no eligible retailers in the LBD. Conversely, it is labeled PR if the destination ZIP contains eligible retailers but no eligible wholesalers. This rule yields about 17,000 labeled shipments. I partition these labeled shipments 90–10 into an initial training set and a validation set. On the 90 percent training set, I use ten-fold cross-validation to compare candidate random-forest specifications (Breiman, 2001); I then use the 10 percent validation set to select the hyperparameter specification. After selecting the specification, I refit the final random forest on all labeled shipments and apply it to processor-origin shipments whose destination ZIP codes contain both downstream types. The predictors include shipment weight, shipment value, shipment distance, survey year, shipper NAICS, SCTG code, frozen status, transportation mode, an intrastate indicator, origin state, and destination state.

The classifier is used as a data-construction tool rather than a standalone economic model. Its purpose is to assign ambiguous processor-origin shipments to economically meaningful downstream channels. Following the distinction emphasized by Mullainathan and Spiess (2017), this step is a supervised prediction and measurement exercise: the random forest constructs the shipment-channel variable, while the structural parameters are estimated only after this variable has been assigned. The rounded confusion matrix in Table A1 reports fitted classifications from the final random forest after it has been refit on the full labeled sample. It is therefore not a held-out confusion matrix. The matrix implies approximately 99 percent fitted accuracy and a true-positive rate for PW links of about 92 percent. Accuracy on the 10 percent validation set used during specification selection was about 96 percent, and the other validation-set classification metrics were comparable. The validation cases are shipments to mechanically separable destination ZIP codes; the true labels for ambiguous ZIPs are not observed directly. Applying the fitted classifier to ambiguous destinations therefore assumes that the mapping from observable shipment characteristics to the PR–PW channel is stable between mechanically separable and mixed ZIP codes. The matrix reports approximate, disclosure-rounded counts and is not intended to reconcile exactly with the approximate labeled-shipment total. Wholesaler-origin shipments are assigned to wholesaler-to-retailer or downstream-distribution links, denoted WR, by construction.

The PR–PW classification is aggregated to the product-market level to construct ToW , which measures the importance of the wholesale channel in a given product-market cell. Specifically, ToW_{jm} is defined as the share of CFS-expanded processor-origin shipment weight

in product j and market m that is classified as processor-to-wholesaler:

$$\text{ToW}_{jm} = \frac{\sum_{s \in (j,m)} \tilde{w}_s \mathbf{1}\{s \in PW\}}{\sum_{s \in (j,m)} \tilde{w}_s}. \quad (1)$$

Here, $\tilde{w}_s$ is the CFS-expanded shipment weight for shipment s . The denominator sums over all processor-origin shipments in the product-market cell, while the numerator restricts the sum to PW shipments. Higher values of ToW_{jm} indicate that a product-market is served more heavily through wholesale intermediaries, whereas lower values indicate greater reliance on direct processor-to-retailer shipments.

In the demand model, ToW_{jm} enters mean utility directly and through an interaction with log distance. The direct term captures differences in downstream buyer composition across markets. The distance interaction allows wholesaler-oriented shipment flows to face a different distance penalty than direct processor-to-retailer flows, reflecting the role of wholesalers in consolidation, inventory holding, and distribution-route networks.

3.3 Poultry Shipment Network

Before estimating the structural model, I construct a descriptive ZIP-code poultry shipment network from CFS shipment records and LBD establishment information. The network documents the spatial organization of poultry flows and summarizes the PR, PW, and WR channels. It is not the maintained BLP market definition; the structural model aggregates processor-origin products to destination-county-year procurement markets.

The variable carried from the network into the structural model is the product-market wholesale-channel share ToW_{jm} from (1); the product-market shipment distance d_{jm} is constructed directly from CFS records in Subsection 3.4. Network density, the Herfindahl–Hirschman index (HHI), directed-edge counts, and link-type shipment shares are descriptive statistics only. Appendix B.1 reports the full ZIP-code network construction, and Table A2 reports the disclosure-cleared network summary.

3.4 Products, Markets, Unit Values, and Quantities

The structural estimation retains shipments with SCTG code 05121, poultry fresh or chilled, and SCTG code 05122, poultry frozen. A product j is an establishment–commodity pair: an establishment interacted with one of these two SCTG codes. A market is a destination-county-year cell, indexed by m , with choice set $\mathcal{J}_m$. This definition treats a plant shipping fresh poultry and the same plant shipping frozen poultry as two separate products while preserving the plant identity needed for ownership and cost controls. Only processor-origin

shipments enter the estimation sample; wholesaler-origin (WR) shipments appear in the descriptive network but not in the structural choice set, and their volume is part of the residual outside option defined in Subsection 3.5.

Shipment weight and CFS sampling weight are distinct variables. For each shipment record, I first multiply the reported shipment weight by the CFS sampling weight and then aggregate within product-market cells. Thus product-market quantity, q_{jm} , is measured as expanded shipment weight,

$$q_{jm} = \sum_{s \in (j,m)} \tilde{w}_s = \sum_{s \in (j,m)} \kappa_s^{\text{CFS}} w_s^{\text{ship}}. \quad (2)$$

As a scale check, I compare the expanded CFS shipment total $\sum_s \kappa_s^{\text{CFS}} w_s^{\text{ship}}$ for the retained poultry scope with the USDA production benchmark for the matched survey years and poultry classes. The retained expanded CFS total is about 85 percent of that benchmark (USDA NASS, various years). This comparison is used only to assess scale: the model quantities remain survey-expanded observed shipments, not a complete production census.

The CFS instructs respondents to report shipment value net of freight charges and excise taxes, on a free-on-board (FOB) plant basis (U.S. Census Bureau, 2017). Accordingly, the product-market FOB-plant unit value is the expanded-value-to-expanded-weight ratio,

$$p_{jm} = \frac{\sum_{s \in (j,m)} \tilde{v}_s}{\sum_{s \in (j,m)} \tilde{w}_s}. \quad (3)$$

Equivalently, this is $\sum_s \kappa_s^{\text{CFS}} v_s / \sum_s \kappa_s^{\text{CFS}} w_s^{\text{ship}}$ within the product-market cell. I refer to p_{jm} throughout as the FOB-plant unit value. The structural model treats this measure as the product price, but it does not include separately billed freight charges. The CFS sampling weights enter at the shipment level before aggregation, and all monetary variables are converted to constant 2017 dollars before estimation. Shipment distance is aggregated with the same expanded shipment weights,

$$d_{jm} = \frac{\sum_{s \in (j,m)} \text{distance}_s \times \tilde{w}_s}{\sum_{s \in (j,m)} \tilde{w}_s}. \quad (4)$$

The estimation uses $\log(d_{jm})$.

Plant-level product categories are constructed from CMF product-line information rather than from shipment temperature status. The CMF reports establishment output using detailed NAICS product codes. For poultry processing, these codes distinguish young chickens, processed poultry and small-game products containing at least 20 percent poultry or meat, turkeys, hens and fowl, and other poultry and small game (Weng, Vukina, and Zheng, 2015).

In my sample, poultry establishments are highly specialized, so the detailed CMF product code uniquely identifies each establishment’s underlying product category. I assign each establishment to one mutually exclusive plant-level category: broiler, processed poultry, or other poultry. This assignment is the source of the broiler and processed-poultry dummy variables used in the demand and supply equations. An establishment can still ship both fresh or chilled and frozen forms of its underlying product. The SCTG code determines only the frozen indicator: code 05122 is frozen and code 05121 is fresh or chilled. Additional plant-level controls include production employment, asset stock, hourly production-worker pay, capital expenditure rate, materials unit cost, supplemental input unit cost, labor unit cost, inventory-to-assets ratio, and indicators for own-slaughter and external-slaughter processing plants. The three unit-cost variables are reported in constant 2017 dollars per pound. Appendix C provides variable definitions.

3.5 Market Size and Shares

Market size converts CFS-expanded inside shipments into product shares and an outside option. Four objects enter the demand model: observed inside shipments R_m , final market size M_m , product shares s_{jm} , and the outside share s_{0m} . The construction starts from observed CFS inside shipments, uses county population and national poultry availability to form a baseline capacity measure, and then adjusts that scale using downstream retail and wholesale activity from ARTS/AWTS and LBD. Appendix B.2 gives the exact market-size formula and small-area construction.

Observed inside shipments in destination-county-year market m are

$$R_m = \sum_{j \in \mathcal{J}_m} q_{jm}. \quad (5)$$

Here R_m is the total CFS-expanded quantity of retained poultry products shipped into market m . It is not total county poultry consumption and not total poultry sales by all suppliers. I use it as the dependent variable in the auxiliary market-size regression and as a lower bound on final market size. The fitted market-size proxy should therefore be read as a capacity measure for potential poultry procurement in a destination-county-year market, not as a census of realized poultry consumption. Product and outside shares are

$$s_{jm} = \frac{q_{jm}}{M_m}, \quad s_{0m} = 1 - \sum_{j \in \mathcal{J}_m} s_{jm}. \quad (6)$$

The outside option is a residual normalization rather than an observed choice. It can be

interpreted as covering procurement from establishments or sourcing arrangements not included in the estimation sample, including shipments excluded by the sample restrictions in Subsection 3.7 and substitution toward other meat products, although these flows are not separately observed or modeled. Because the market-size proxy is calibrated to downstream meat retail and wholesale capacity rather than to poultry shipments alone, it deliberately leaves room for this residual, and the counterfactual welfare comparisons hold the resulting normalization fixed.

3.6 Market-Specific Buyer-Type Distribution

The economic motivation is that food-service, retail, and wholesale buyers may not respond to price changes in the same way. Restaurants, grocery retailers, and wholesale distributors differ in scale, inventory needs, and substitution opportunities. I therefore allow price sensitivity to vary with the downstream buyer type represented in each destination market.

The difficulty is that the CFS records the shipment destination but not the receiving establishment. I use the LBD to approximate the local buyer mix. In each destination-county-year market m , I identify eligible downstream establishments and classify them as food service, retail, or wholesale. Let $\mathcal{A}_m$ be this set of establishments, and let $T_{am} \in \{F, R, W\}$ denote the type of establishment a . To convert local establishments into buyer-type demand weights, I combine establishment employment, emp_{am} , with a buyer-type throughput-per-employee factor, $\tau_{T_{am}}$. The resulting market-specific buyer-type distribution is

$$\phi_{gm} = \frac{\sum_{a \in \mathcal{A}_m} \text{emp}_{am} \tau_{T_{am}} \mathbf{1}\{T_{am} = g\}}{\sum_{a \in \mathcal{A}_m} \text{emp}_{am} \tau_{T_{am}}}, \quad g \in \{F, R, W\}. \quad (7)$$

Conceptually, ϕ_{gm} is the constructed probability that one representative unit of poultry demand in market m is associated with buyer type g . A high value of ϕ_{Wm} means that the market has a large wholesale presence after accounting for both employment and poultry throughput per worker. These probabilities are not directly observed shipment-level buyer identities; they are an empirical proxy built from local business records and the throughput calibration.

For estimation, I represent this distribution with 1,000 representative buyer records drawn with replacement. The probability of drawing establishment a is proportional to $\text{emp}_{am} \tau_{T_{am}}$, so the simulated shares of food-service, retail, and wholesale records approximate ϕ_{Fm} , ϕ_{Rm} , and ϕ_{Wm} . These records are representative LBD establishment records, not household consumers and not an additional survey. Appendix B.3 gives the NAICS mapping, throughput calibration, and sampling equations.

The estimation file keeps these representative buyer records so that the demand specification can use the local buyer-type mix. For each representative buyer record, I retain the market identifier and two buyer-type indicators,

$$D_{im}^R = \mathbf{1}\{T_{im} = R\}, \quad D_{im}^W = \mathbf{1}\{T_{im} = W\}. \quad (8)$$

Here retail and wholesale records are measured relative to food service, the omitted buyer type. Section 4 shows how these indicators allow price sensitivity to differ across buyer types. I do not include establishment employment or payroll as separate buyer characteristics because employment already affects the representative-buyer sampling probability, and payroll is mechanically related to employment.

3.7 Sample Restrictions and Descriptive Statistics

The estimation sample excludes establishments with internal plant transfers greater than 30 percent of total shipment value, trims product-market observations outside the 5th and 95th percentiles of the FOB-plant unit-value distribution, drops product-market observations with shares below 0.1 percent, and removes markets with fewer than four inside products. Shipment volume excluded by these restrictions is not reassigned to retained products; under the market-size construction in Subsection 3.5, it falls into the residual outside option as a matter of normalization rather than as an observed reallocation. The final sample contains 700 destination-county-year markets, 6,500 product-market observations, 400 establishment–commodity products, and 700,000 representative buyer records.

Table A3 reports descriptive statistics for the estimation variables. The average FOB-plant unit value is 1.423 dollars per pound (\$/lb), and the average product share is 0.046. The mean log distance is 6.09, with substantial dispersion across product-market cells. The sample is concentrated in broiler products but still contains meaningful frozen and processed-poultry variation: broiler, frozen, and processed-poultry indicators have means of 0.600, 0.277, and 0.246, respectively. The mean wholesale-channel share ToW is 0.093, indicating that most processor-origin shipment weight is classified as direct-to-retailer, although the standard deviation of 0.282 shows that some product-market cells rely much more heavily on wholesale intermediaries.

Because the CFS is a sample of shipments rather than a census, each product-market cell aggregates a finite number of sampled shipment records, and the choice set in each market is defined by sampled positive flows. Table A4 reports the resulting sampling depth. Thin cells are common but economically small: cells built from one or two sampled shipments account for 29.5 percent of observations but only 4.5 percent of CFS-expanded shipment

weight, cells with five or more shipments carry 83 percent of expanded weight, and the weight-weighted median cell is built from about 12 shipments, so sampling noise in shares and unit values is concentrated where the welfare stakes are smallest. Defining availability by sampled positive flows also means that products a buyer could in principle purchase can be absent from the estimated choice set, and cells that survive the share restrictions are selected partly on realized demand, a selection issue emphasized by [Gandhi, Lu, and Shi \(2023\)](#). Unsourced and excluded flows fall into the residual outside option as a matter of normalization (Subsection 3.5), and the estimates should be read as conditional on the sampled choice sets.

4 Model and Estimation

The structural model is based on the shipment-network variables and market-specific buyer-type distribution constructed in Section 3. The channel classification produces product-market variables such as ToW; the LBD sampling procedure produces representative buyer records that approximate the local food-service, retail, and wholesale mix. Conditional on these objects, the model estimates demand, recovers marginal costs from firms’ pricing first-order conditions, and supplies the primitives for the counterfactuals.

4.1 Demand

Demand is modeled using a random-coefficients nested logit model with the market-specific buyer-type distribution represented by LBD representative buyer records. For representative buyer record i , product j , and destination-county-year market m , indirect utility is

$$u_{ijm} = \delta_{jm} + \mu_{ijm} + \varepsilon_{ijm}, \quad (9)$$

where ε_{ijm} has the nested extreme-value structure. Mean utility is

$$\delta_{jm} = x'_{jm}\beta + \xi_{jm}, \quad (10)$$

where x_{jm} collects observed product-market demand characteristics, β is the corresponding vector of mean-utility coefficients, and ξ_{jm} is unobserved demand quality. The buyer-specific component is

$$\mu_{ijm} = -\alpha_{im}p_{jm}. \quad (11)$$

Here p_{jm} is the FOB-plant unit value defined in Subsection 3.4, which the structural model treats as the product price. The price-sensitivity parameter is positive and lognormally

distributed conditional on buyer type:

$$\log \alpha_{im} = \pi_F + \pi_R D_{im}^R + \pi_W D_{im}^W + \sigma v_{im}, \quad v_{im} \sim \mathcal{N}(0, 1). \quad (12)$$

Equivalently, $\alpha_{im} = \exp(\pi_F + \pi_R D_{im}^R + \pi_W D_{im}^W + \sigma v_{im})$, so price has a negative utility effect for every simulated buyer. The food-service type is omitted, so π_F is the food-service baseline log price sensitivity, while π_R and π_W measure retailer and wholesaler differences relative to food service. Beyond enriching substitution patterns, this price-coefficient heterogeneity matters for the counterfactuals: multinomial and nested logit constrain demand to be log-concave, which mechanically caps cost pass-through below one, whereas random coefficients on price let the demand curvature that governs pass-through and merger effects be determined by the data (Miravete, Seim, and Thurk, 2023).

Products are partitioned into three nests, $\mathcal{G} = \{B, P, O\}$, using broiler, processed-poultry, and other-poultry product categories:

$$g(j) = \begin{cases} B, & \text{broiler}_{jm} = 1, \\ P, & \text{processed}_{jm} = 1, \\ O, & \text{broiler}_{jm} = 0 \text{ and } \text{processed}_{jm} = 0. \end{cases} \quad (13)$$

The frozen indicator remains an observed product characteristic in mean utility rather than a nest definition. Let $\mathcal{J}_{g,m}$ denote products in nest g in market m , and let $\rho \in [0, 1)$ be the nesting parameter. With the outside option normalized to zero, define the agent-specific inclusive value

$$I_{igm}(\theta) = \sum_{k \in \mathcal{J}_{g,m}} \exp \left(\frac{\delta_{km} + \mu_{ikm}}{1 - \rho} \right). \quad (14)$$

Agent i 's probability of choosing product j is the product of the conditional within-nest probability and the nest probability:

$$P_{ij|g(j),m}(\theta) = \frac{\exp((\delta_{jm} + \mu_{ijm})/(1 - \rho))}{I_{ig(j),m}(\theta)}, \quad (15)$$

$$P_{ig,m}(\theta) = \frac{I_{igm}(\theta)^{1-\rho}}{1 + \sum_{h \in \mathcal{G}} I_{ihm}(\theta)^{1-\rho}}, \quad (16)$$

$$P_{ijm}(\theta) = P_{ij|g(j),m}(\theta) P_{ig(j),m}(\theta). \quad (17)$$

When $\rho = 0$, these probabilities collapse to the standard random-coefficients logit probabilities. Positive ρ allows substitution to be stronger within the broiler, processed-poultry, and other-poultry nests than across nests. Model-implied product shares average these probabil-

ities over the representative LBD buyer records in each market:

$$s_{jm}(\theta) = \sum_{i=1}^{1,000} \psi_{im} P_{ijm}(\theta), \quad \psi_{im} = \frac{1}{1,000}. \quad (18)$$

Observed shares are inverted into mean utilities using the BLP contraction mapping (Berry, 1994; Berry, Levinsohn, and Pakes, 1995).

In the empirical specification, x_{jm} includes the frozen-shipment indicator, plant product-category indicators, the private-carriage share, the wholesale-channel share ToW, log shipment distance, interactions between log distance and the product or shipment attributes, and survey-year indicators. This specification lets geography and the predicted wholesale-channel share enter demand directly, and it lets the distance penalty differ by product and shipment type. Buyer type does not enter δ_{jm} because it is an agent characteristic rather than a product-market characteristic; instead, buyer type shifts the price coefficient through (12).

4.2 Supply

The model treats the observed FOB-plant unit value as the product price set in a static Bertrand–Nash equilibrium. Let $\mathcal{H}_{jk,m} = 1$ if products j and k in market m are owned by the same firm and zero otherwise. The market-level first-order conditions can be written as

$$\mathbf{p}_m = \mathbf{m}\mathbf{c}_m + \Delta_m^{-1} s_m(\theta), \quad (19)$$

where

$$\Delta_{jk,m} = -\mathcal{H}_{jk,m} \frac{\partial s_{km}(\theta)}{\partial p_{jm}}. \quad (20)$$

Ownership is measured with the LBD firm identifier, and the implied markup for product j is $p_{jm} - mc_{jm}$.

Marginal cost is parameterized in levels with a linear regression,

$$mc_{jm} = w'_{jm} \gamma + \omega_{jm}, \quad (21)$$

where w_{jm} includes plant-level unit costs, input-mix variables, shipment characteristics, plant-type indicators, year indicators, destination population, and rival-count controls. This specification relates recovered marginal cost in levels to supply-side shifters; its parameters are estimated jointly with the demand parameters through the supply-side moment condition below.

4.3 Instruments, Identification, and Estimation

FOB-plant unit values are endogenous because unobserved product quality and cost shocks may be correlated with the observed unit value. The excluded demand instruments combine plant-level cost and scale shifters, market-structure measures, and local differentiation instruments. Appendix B.4 lists the variables in the instrument set. Plant cost and scale shifters move unit values through supply-side cost and capacity conditions, while local differentiation instruments shift markups through exogenous crowding in product-characteristic space. These instruments are valid under the maintained assumption that, after the observed controls and market construction are conditioned on, they are not correlated with unobserved demand quality. The local differentiation instruments are constructed from frozen status, distance, broiler status, turkey status, and processed-poultry status, following the logic that exogenous crowding in characteristic space shifts substitution patterns and pricing incentives (Gandhi and Houde, 2019). The efficiency of random-coefficients demand estimates also depends on the choice of excluded instruments (Reynaert and Verboven, 2014); I use a fixed set of excluded instruments rather than iterating to approximate optimal instruments.

Shipment distance therefore has two roles in the empirical specification. The product's own log distance enters mean utility as an included, conditionally exogenous product characteristic and serves as its own instrument. The differentiation instruments built from distance, frozen status, and product category are excluded instruments for the endogenous FOB-plant unit value. The own-distance coefficient is identified under the maintained condition $\mathbb{E}[\log(d_{jm})\xi_{jm}] = 0$: after conditioning on observed controls and market construction, shipment distance is uncorrelated with unobserved demand quality.

The identifying assumptions are the standard BLP orthogonality conditions,

$$\mathbb{E}[z_{jm}^d \xi_{jm}(\theta)] = 0, \quad (22)$$

$$\mathbb{E}[z_{jm}^s \omega_{jm}(\theta, \gamma)] = 0, \quad (23)$$

where z_{jm}^d and z_{jm}^s denote demand- and supply-side instruments. Here $\theta = (\beta', \pi_F, \pi_R, \pi_W, \sigma, \rho)'$ collects the demand-side parameters, and γ is the cost-coefficient vector in (21); the supply residual evaluates (21) at the marginal costs recovered from the first-order conditions (19). These assumptions require the excluded instruments to shift the FOB-plant unit value or cost without being correlated with unobserved demand quality, after conditioning on observed controls and market construction. Product-count instruments are therefore interpreted conditional on the maintained market definition and sample restrictions, not as a model of endogenous entry.

The parameters are estimated by two-step GMM ([Hansen, 1982](#)),

$$(\hat{\theta}, \hat{\gamma}) = \arg \min_{\theta, \gamma} \hat{g}(\theta, \gamma)' \Omega \hat{g}(\theta, \gamma), \quad (24)$$

where $\hat{g}(\theta, \gamma)$ stacks the demand- and supply-side sample moments and Ω is the two-step GMM weighting matrix, so the demand and cost parameters are estimated jointly. The numerical implementation uses L-BFGS-B optimization, 1,000 representative LBD buyer records per market, and `pyblp` routines for BLP estimation and post-estimation analysis ([Conlon and Gortmaker, 2020](#)). Each representative record contains the buyer-type indicators in Subsection 3.6 and one standard-normal simulation node for the lognormal price coefficient. Standard errors clustered by destination county are reported throughout; clustering at the county level allows for correlation across products within a market and across survey years within the same county, including the market-level variation in the rival-count instruments. Instrument relevance is supported by released weak-instrument diagnostics in addition to the OLS–IV unit-value contrast in Section 5. In the linear benchmarks, the cluster-robust Montiel Olea–Pflueger effective first-stage F statistic is 77.62 for the IV logit, far above the critical value of about 23 associated with a worst-case bias of 10 percent ([Montiel Olea and Pflueger, 2013](#)). For the nested-logit specification, which instruments both the unit value and the within-nest share, the Kleibergen–Paap rk Wald F statistic is 58.19 ([Kleibergen and Paap, 2006](#)); because formal critical values for robust statistics with multiple endogenous variables are not established, this statistic is reported as a descriptive diagnostic. Disclosure constraints limit other auxiliary estimation output, such as full first-stage coefficient vectors.

5 Results

This section reports the estimates in the order used to build the empirical model: preliminary logit benchmarks, BLP demand estimates with LBD buyer-type variation, and recovered supply-side cost parameters and economic magnitudes. All estimates use the estimation sample described in Subsection 3.7.

5.1 Preliminary Logit Benchmarks

Before estimating the random-coefficients model, I use three logit-based specifications as diagnostic benchmarks. They are not the final welfare model, but they show whether the main ingredients of the structural specification are visible in simpler share equations. Following

the standard market-share inversion in [Berry \(1994\)](#), define

$$y_{jm} = \log(s_{jm}) - \log(s_{0m}), \quad (25)$$

where s_{jm} is the share of establishment–commodity product j in market m , and s_{0m} is the outside option share. Thus y_{jm} is the product’s mean utility relative to the outside option.

The OLS and IV logit benchmarks estimate

$$y_{jm} = x'_{jm}\beta + \alpha p_{jm} + \xi_{jm}, \quad (26)$$

where p_{jm} is the FOB-plant unit value and x_{jm} contains the same observed demand controls as in the BLP specification: the frozen-shipment indicator, plant product-category indicators, the private-carriage share, the wholesale-channel share ToW, log shipment distance, distance interactions, and survey-year indicators. The IV version instruments the unit value with the excluded demand instruments described in [Section 4](#).

The third benchmark is a nested logit model,

$$y_{jm} = x'_{jm}\beta + \alpha p_{jm} + \rho \log(s_{j|g(j),m}) + \xi_{jm}, \quad (27)$$

where $s_{j|g(j),m}$ is the share of product j within the broiler, processed-poultry, or other-poultry nest defined in [\(13\)](#). The frozen indicator remains an observed product characteristic but does not define the nest. The parameter ρ summarizes within-nest substitution. The nested-logit benchmark is estimated by IV with the same excluded instrument set as the IV logit, instrumenting both the FOB-plant unit value and the within-nest share.

[Table 1](#) reports the detailed benchmark estimates. Three patterns motivate the structural specification. First, the coefficient on the FOB-plant unit value is negative in all columns and becomes more negative after instrumenting the unit value, moving from -1.062 in OLS to -1.898 in IV. This pattern is consistent with the standard concern that unit values are correlated with favorable unobserved demand shocks. The excluded instruments are strong: the cluster-robust Montiel Olea–Pflueger effective first-stage F statistic is 77.62 for the IV logit, far above its critical value of about 23, while the Kleibergen–Paap rk Wald F statistic for the nested-logit specification, which instruments both the unit value and the within-nest share, is 58.19, reported as a descriptive diagnostic because formal critical values for robust statistics with multiple endogenous variables are not established ([Montiel Olea and Pflueger, 2013](#); [Kleibergen and Paap, 2006](#)). Second, log shipment distance is negative and precisely estimated across specifications. Conditional on the FOB-plant unit value and observed product attributes, shipments from more distant plants receive lower market shares relative

to the outside option, so geography enters demand as a non-price procurement friction as well as a cost shifter.

The distance interactions are useful mainly as diagnostics. The positive broiler-by-distance interaction implies a weaker distance penalty for broiler products than for the omitted product group. The ToW interaction is negative in the OLS and IV specifications, suggesting that product-market cells with a higher wholesale-channel share are more sensitive to distance, but this pattern becomes smaller and statistically insignificant once within-nest substitution is included.

Third, the nested-logit estimate of ρ is 0.448 with a standard error of 0.022. This positive and precise estimate indicates stronger substitution within the broiler, processed-poultry, and other-poultry nests than across nests. The preliminary estimates therefore support treating the FOB-plant unit value as endogenous, keeping shipment distance in mean utility, and using a richer random-coefficients nested logit model for the main estimates and welfare simulations.

5.2 BLP Demand Estimates

Table 2 reports the nonlinear price-sensitivity parameters. The baseline food-service log price-sensitivity parameter π_F is 1.138 and is precisely estimated. The retailer increment π_R is 0.686, and the wholesaler increment π_W is 1.202; both are positive and statistically significant at the 1 percent level. Under the lognormal specification, holding the idiosyncratic normal draw fixed, retailer agents have price sensitivity about $\exp(0.686) = 1.99$ times the food-service baseline, while wholesaler agents have price sensitivity about $\exp(1.202) = 3.33$ times the food-service baseline. The wholesaler effect is the largest buyer-type difference, and it matches the pricing institutions described in Subsection 2.2: merchant wholesalers resell near-commodity poultry on thin operating margins and monitor wholesale quotations across many competing suppliers, so a given unit-value increase cuts far deeper into their resale margin than it does for a food-service buyer whose purchases are specification- and menu-constrained. The within-type dispersion parameter σ , 0.364, shows that meaningful residual heterogeneity remains even after conditioning on buyer type.

The LBD buyer data play two distinct roles. The sampling procedure in Appendix B.3 determines how much probability mass food-service, retail, and wholesale establishments receive in each market-specific buyer-type distribution. Table 2 then estimates how the price coefficient differs across the sampled buyer types. Thus the coefficients π_R and π_W are not product-level demand shifters and should not be read as reduced-form effects of selling to retailers or wholesalers. They are demographic interactions in the random-coefficients demand

Table 1: Logit, IV Logit, and Nested-Logit Demand Estimates

Variable	Logit	IV logit	Nested logit
FOB-plant unit value	-1.0620*** (0.0823)	-1.8980*** (0.2929)	-2.7150*** (0.2821)
Constant	-0.8081** (0.3262)	0.1142 (0.5095)	1.7420*** (0.4941)
Frozen-shipment indicator	0.0954 (0.2446)	0.2397 (0.2469)	0.0136 (0.2287)
Broiler product indicator	-0.7386** (0.3156)	-0.5744* (0.3147)	-0.3506 (0.3046)
Processed-poultry indicator	-0.3529 (0.3492)	-0.2270 (0.3477)	0.1925 (0.3337)
private	-0.1365 (0.2471)	-0.1248 (0.2484)	-0.1786 (0.2209)
ToW	0.9309** (0.4132)	1.1370*** (0.4280)	0.6115 (0.3963)
Log shipment distance	-0.2131*** (0.0469)	-0.1851*** (0.0466)	-0.1956*** (0.0453)
Frozen $\times$ log distance	-0.0219 (0.0386)	-0.0355 (0.0388)	-0.0390 (0.0359)
Broiler $\times$ log distance	0.1839*** (0.0491)	0.1550*** (0.0491)	0.2184*** (0.0475)
Processed poultry $\times$ log distance	0.0741 (0.0546)	0.0615 (0.0547)	0.0646 (0.0523)
private $\times$ log distance	0.0264 (0.0431)	0.0247 (0.0434)	0.0369 (0.0384)
ToW $\times$ log distance	-0.1184* (0.0647)	-0.1439** (0.0669)	-0.0685 (0.0617)
2012 indicator	0.6168*** (0.0539)	0.6914*** (0.0544)	0.5170*** (0.0493)
2017 indicator	0.1569*** (0.0397)	0.1874*** (0.0415)	0.1976*** (0.0380)
Nested-logit parameter (ρ)	— —	— —	0.4480*** (0.0216)

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. The IV logit model instruments the FOB-plant unit value with the excluded demand instruments used in the BLP model. The nested-logit column is also estimated by IV with the same excluded instrument set, instrumenting the FOB-plant unit value and the within-nest share; it uses broiler, processed-poultry, and other-poultry nests, and frozen shipment status is an observed control rather than a nest. The cluster-robust Montiel Olea–Pflueger effective first-stage F statistic is 77.62 in the IV logit column; the Kleibergen–Paap rk Wald F statistic is 58.19 in the nested-logit column. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table 2: Random-Coefficients Price Sensitivity by Buyer Type

Parameter	Interpretation	Estimate
π_F	Food-service baseline log price sensitivity	1.1380*** (0.1244)
π_R	Retailer increment relative to food service	0.6862*** (0.2195)
π_W	Wholesaler increment relative to food service	1.2020*** (0.1109)
σ	Within-type lognormal dispersion	0.3644*** (0.0441)

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. Parameter notation follows (12). Food-service agents are the omitted buyer type. Positive retailer and wholesaler increments raise log price sensitivity relative to food-service agents. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

system: they shift the price sensitivity of retailer and wholesaler agents relative to food-service agents. Because the CFS does not observe receiver-level choices, these coefficients are model-based heterogeneity: they are identified from the covariation between market demand responses and the calibrated buyer-type distribution, conditional on the throughput calibration in Appendix B.3.

Table A5 reports the mean-utility parameters, holding fixed the same observed demand controls used in the logit benchmarks. Distance enters negatively and precisely, with a coefficient of -0.210 , confirming that shipments from more distant plants have lower mean utility even after the FOB-plant unit value and product characteristics are controlled for. The broiler-by-distance interaction is positive and statistically significant; its magnitude nearly matches the distance coefficient, a pattern taken up at the end of this subsection. The wholesale-channel share ToW has a positive level effect, partly offset by a negative distance interaction. The table also reports the BLP nesting parameter ρ , estimated at 0.323 with a standard error of 0.034 . As in the preliminary nested logit, this parameter is based on broiler, processed-poultry, and other-poultry nests. It belongs with the demand-side estimates because it governs within-nest substitution under this product-category partition, not supply-side marginal cost. The mean-utility estimates therefore preserve the geographic demand pattern seen in Table 1, while Table 2 adds a second dimension of heterogeneity through the local LBD buyer-type distribution.

The distance coefficients reveal that geography operates through different channels for commodity and differentiated products. For the omitted other-poultry category, log distance

carries the full mean-utility penalty of -0.210 ; for processed poultry the positive interaction offsets part of it; and for broiler products the interaction ($+0.216$) offsets the penalty almost exactly, so conditional on the FOB-plant unit value, broiler demand is essentially insensitive to distance. This pattern is economically coherent rather than anomalous. Broilers are the closest product in the sample to a commodity, so buyers substitute freely across origins once the unit value is accounted for, and distance reaches the broiler segment through the supply side, where it raises marginal cost (Table A6) and hence equilibrium unit values. Differentiated products face an additional non-price distance friction, consistent with the reliability, freshness, and service considerations that price alone cannot arbitrage. Geography therefore matters for every product in the sample, but through the cost channel for the commodity segment and through both channels for differentiated products.

5.3 Supply-Side Estimates

Table A6 reports the supply-side cost parameters from (21), where the dependent variable is the marginal cost recovered in levels from the multi-product Bertrand first-order conditions in (19). An external check is worth stating before the coefficients. The recovered marginal costs are anchored by demand-side pricing behavior alone—each is a unit value minus a model-implied markup—yet their average of 0.966 \$/lb lies within about 4 percent of the sum of the independently observed accounting unit costs for materials, supplemental inputs, and labor ($0.686 + 0.168 + 0.150 \approx 1.00$ \$/lb in Table A3). Exact equality is not expected, since marginal cost need not equal average accounting cost, but the demand system recovers a cost level that the manufacturing records corroborate, and this agreement disciplines the 33 percent average margin before any cost-side coefficient is interpreted. The estimates indicate that frozen shipments and longer shipment distance are associated with higher marginal cost. The frozen indicator raises recovered marginal cost by 0.082 \$/lb, about 8 percent of the average recovered cost of 0.966 \$/lb, consistent with the additional freezing, storage, and cold-chain requirements of frozen shipments. Log shipment distance enters at 0.028 \$/lb per log point, so roughly tripling shipment distance is associated with about a 3 percent increase in average marginal cost. Because the FOB-plant unit value excludes separately billed freight charges, this distance gradient reflects distance-related costs borne at the plant, such as packaging, cold-chain preparation, and shipper-provided carriage, rather than billed freight. The processed-poultry level effect is negative, while the processed-by-distance interaction is positive, so the net cost implication for processed products varies with distance; at the sample mean log distance of about 6.1, the two terms approximately offset.

Plant-level unit costs for materials, supplemental inputs, and labor enter positively and

precisely, as do capital expenditure and inventories, so plants with higher observed input costs per pound are recovered as higher-marginal-cost suppliers. The unit-cost variables are measured in constant 2017 dollars per pound. The slaughter-type indicators are also informative about plant organization: external-slaughter plants, which process poultry slaughtered elsewhere, have recovered marginal costs about 0.108 \$/lb higher than the omitted plant group, while own-slaughter plants show a small and only marginally significant negative differential, a pattern possibly consistent with integrated slaughter economizing on purchased-input costs. These coefficients summarize how recovered marginal costs covary with observed cost shifters in the estimation sample; they should not be read as an accounting decomposition of marginal cost or as a mechanical pass-through equation. Rival-count variables should be interpreted as conditional cost or market-condition controls, not as standalone causal effects of rivals on costs.

5.4 Economic Magnitudes

The preliminary and BLP estimates deliver a coherent set of economic magnitudes. The logit benchmarks show that unit-value endogeneity matters and that substitution is stronger within the broiler, processed-poultry, and other-poultry nests than across nests. The random-coefficients nested logit adds local buyer composition by allowing price sensitivity to vary with the LBD buyer-type distribution. The estimated price-sensitivity interactions imply that a county-year market with greater wholesale-buyer mass can have different pass-through and substitution responses than an otherwise similar market dominated by food-service establishments. The supply estimates then translate these substitution patterns into markups through the multi-product ownership matrix.

These magnitudes provide the bridge to the counterfactual analysis. Average own-price elasticity is elastic,¹ the average recovered unit-value–cost margin is economically meaningful, and the distance variables appear both in demand and cost. Consequently, the counterfactuals below are not reduced-form scenario exercises: they recompute equilibrium FOB-plant unit values, quantities, markups, and welfare using the estimated substitution patterns, the recovered marginal-cost function, and the relevant ownership matrix. This is why similar aggregate total-surplus losses can have different economic interpretations depending on

¹The elasticities here are product-level objects—establishment–commodity products within destination-county markets—and are naturally larger in magnitude than category-level poultry demand elasticities, which are typically inelastic, because buyers substitute across competing suppliers within the category. The same disaggregation pattern appears elsewhere: within FSRDC poultry research, plant-level broiler demand is estimated at about -1.3 against aggregate chicken elasticities of -0.3 to -1.1 (Weng, Vukina, and Zheng, 2015), and in distilled spirits, category demand is inelastic while product-level own-price elasticities range from about -2.5 to -4.5 (Conlon and Rao, 2020).

whether the primitive shock removes products, raises logistics costs, raises raw-poultry costs, or changes ownership incentives.

6 Counterfactual Simulations

The counterfactuals translate the estimated model into four supply-chain scenarios. The descriptive network and representative-buyer construction define the baseline demand environment; the BLP and supply-side estimates recover substitution, marginal costs, and markups; the scenarios change one primitive at a time and recompute equilibrium outcomes. The four counterfactual scenarios isolate distinct mechanisms: plant unavailability in confidentially selected counties, a logistics shock that raises effective shipment distance, an upstream input-cost shock that raises raw-poultry costs, and an ownership shock that merges the third- and fourth-largest firms.

6.1 Benchmark and Welfare Aggregation

The counterfactual analysis is anchored to benchmark equilibrium objects recovered from the estimated demand and supply system. These objects summarize the model-implied baseline before any counterfactual shock is introduced: FOB-plant unit values, recovered marginal costs, elasticities, inside sales, producer surplus, downstream-buyer surplus, and total surplus. The baseline row of [Table 3](#) reports these aggregates. The relevant comparison is not the absolute level of surplus by itself, but how the same equilibrium objects change when one primitive changes and all estimated behavioral parameters are held fixed.

Own- and cross-price elasticities are computed as

$$\epsilon_{jk,m} = \frac{\partial s_{jm}(\hat{\theta})}{\partial p_{km}} \frac{p_{km}}{s_{jm}}. \quad (28)$$

The own-price terms summarize the local demand response to a product’s own FOB-plant unit value, while the off-diagonal terms govern the degree of substitution toward rival establishment–commodity pairs in the same destination-county-year market. In the model-implied baseline, the average state-level own-price elasticity is -3.135 . This value implies elastic procurement demand at observed unit values, so the recovered pricing equations attribute positive markups to firms while still disciplining those markups through substantial buyer substitution. The state rows in [Table A7](#) show that this average masks meaningful geographic heterogeneity: baseline elasticities are more negative in some destination states and closer to unit elastic in others, reflecting differences in market size, product availability,

Table 3: Aggregate Counterfactual Outcomes by Scenario

Scenario	FOB-plant unit value	Cost	Elas.	Inside	Producer	Buyer	Total	ΔTS	ΔTS (%)
Model-implied baseline	1.443	0.966	-3.135	63.56	34.39	55.36	89.75	0.00	0.00
Plant unavailability	1.448	0.967	-3.118	61.08	33.04	52.38	85.43	-4.32	-4.81
$\log(\text{distance}) \times 1.2$	1.469	0.990	-3.158	60.98	33.18	52.41	85.60	-4.15	-4.62
Raw-poultry cost +20%	1.558	1.085	-3.515	54.27	27.28	46.58	73.85	-15.90	-17.72
Merger of third- and fourth-largest firms	1.542	0.966	-3.276	57.70	37.09	48.83	85.92	-3.83	-4.27

Notes: The FOB-plant unit value and cost are simple averages across the 36 state rows and are reported in \$/lb; elasticity is unitless and is averaged across state rows. Inside sales are summed across state rows and reported in billion lb. Producer surplus, buyer surplus, total surplus, and ΔTS are reported in billions of dollars. Producer and buyer surplus may not sum exactly to total surplus because each entry is rounded separately from the underlying state rows. Surplus levels are reported under the baseline market-size normalization, and scenario differences hold that normalization fixed. ΔTS is the change in total surplus relative to the model-implied baseline. In the merger scenario, firms are ranked by total CFS-expanded shipment value in the estimation sample. Counterfactual outcomes are model-implied point values evaluated at the estimated parameters; see Appendix E.

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plant proximity, and the composition of poultry products shipped into the state. [Figure A1](#) maps the corresponding baseline state-level distributions.

The baseline average FOB-plant unit value is 1.443 \$/lb, and the recovered average marginal cost is 0.966 \$/lb. This unit value is a simple average across the disclosure-cleared state rows and therefore differs slightly from the unweighted product-market mean of 1.423 in [Table A3](#). The unit-value–cost difference, 0.477 \$/lb, corresponds to an average margin of roughly 33 percent of the unit value. This margin should be interpreted as the model-implied operating markup for the establishment–commodity bundles in the estimation sample, not as an accounting profit rate for the entire firm. It is nonetheless a useful benchmark because the counterfactuals below separate shocks that mainly change marginal costs from shocks that mainly change pricing incentives. For example, a raw-input-cost shock moves the recovered cost component directly, whereas a merger changes the ownership matrix in the pricing first-order conditions while leaving the cost primitives essentially unchanged.

Aggregate inside sales in the baseline are 63.56 billion lb. Under the baseline market-size normalization, baseline producer surplus is 34.39 billion dollars, downstream-buyer surplus is 55.36 billion dollars, and total surplus is 89.75 billion dollars. These surplus levels are conditional on the market-size normalization in [Appendix B.2](#); because every scenario holds that normalization fixed, the scenario differences reported below are more robust objects than the levels themselves. [Table A8](#) probes this normalization directly: re-estimating the model and re-solving all four scenarios with the outside-option normalization set to 0.6 or 0.8 in place of the baseline 0.7 moves the average elasticity by at most 3 percent and the

welfare levels by at most about 5 percent, because market size and surplus per unit offset; every scenario sign and ranking is unchanged, and each total-surplus loss moves by less than 0.6 percentage points per 0.1 change in the normalization. Thus, downstream buyers receive a larger share of measured surplus in the baseline, while producers retain a sizable share through markups. This decomposition is important for interpreting the merger exercise: a scenario can reduce total surplus even when producer surplus rises, because the buyer-surplus loss and the quantity reduction can dominate the producer gain. Conversely, shocks to logistics or raw-poultry costs tend to reduce both producer and buyer surplus because they raise model-implied unit values or make products less attractive and contract the inside market.

The main comparisons below emphasize changes relative to this common baseline. In [Table 3](#), the FOB-plant unit value and cost are reported in \$/lb, elasticity is unitless, inside sales are reported in billion lb, and all surplus measures are reported in billions of dollars. Unit value, cost, and elasticity are simple averages across the disclosure-cleared state rows; inside sales and surplus are sums across those rows. This convention makes the table a compact aggregate counterpart to the detailed state-level outcomes reported in [Table A7](#).

6.2 Counterfactual Procedure and Shock Design

For counterfactual scenario c , estimated demand and cost parameters are held fixed at $(\hat{\theta}, \hat{\gamma})$ while market primitives are changed. The fitted demand and cost unobservables, $\hat{\xi}_{jm}$ and $\hat{\omega}_{jm}$, are also held at their baseline values for products that remain in the choice set. The new equilibrium solves

$$\mathbf{p}_m^c = \mathbf{m}c_m^c(\hat{\gamma}, \hat{\omega}_m) + \Delta_m^c(\mathbf{p}_m^c; \hat{\theta})^{-1} s_m^c(\mathbf{p}_m^c; \hat{\theta}). \quad (29)$$

The procedure follows standard counterfactual simulation in differentiated-products models ([Nevo, 2000b](#)). As is common in that literature, scenario outcomes are reported as model-implied values evaluated at the estimated parameters; [Appendix E](#) discusses the associated uncertainty caveats. Producer surplus is

$$PS_m^c = \sum_{j \in \mathcal{J}_m^c} (p_{jm}^c - mc_{jm}^c) q_{jm}^c, \quad q_{jm}^c = M_m s_{jm}^c(\mathbf{p}_m^c; \hat{\theta}), \quad (30)$$

where the market size M_m is held at its baseline value, and downstream-buyer surplus is the nested-logit BLP log-sum measure. For each representative buyer record, define the

counterfactual nest inclusive value

$$I_{igm}^c(\hat{\theta}) = \sum_{k \in \mathcal{J}_{g,m}^c} \exp\left(\frac{\delta_{km}^c + \mu_{ikm}^c}{1 - \hat{\rho}}\right), \quad (31)$$

where $\mathcal{J}_{g,m}^c$ is the set of available products in nest g under scenario c . The buyer-surplus measure is

$$BS_m^c = M_m \sum_i \psi_{im} \frac{\log\left(1 + \sum_{g \in \mathcal{G}} I_{igm}^c(\hat{\theta})^{1-\hat{\rho}}\right)}{\hat{\alpha}_{im}}. \quad (32)$$

Total surplus is $TS_m^c = PS_m^c + BS_m^c$.

The logistics scenario scales each product's log shipment distance by 1.2, so $\log d_{jm} \mapsto 1.2 \log d_{jm}$, equivalently $d_{jm} \mapsto d_{jm}^{1.2}$. Because the shock is multiplicative in logs, it should be read as a severe logistics-frictions scenario rather than as a 20 percent increase in miles. The effect is largest for shipments that are already distant.

The raw-poultry input-cost scenario raises the raw-poultry cost term, measured from the CMF materials detail in constant 2017 dollars per pound of CFS-expanded shipment weight, by 20 percent and then resolves the pricing first-order conditions with the other cost components and behavioral parameters fixed.

This one-for-one loading is deliberate, and it is not the cross-sectional materials coefficient in [Table A6](#). The cost regression projects recovered marginal cost on observed plant characteristics; its materials coefficient (0.0885) is identified from cross-plant variation in materials cost per pound, which mixes product composition, slaughter arrangements, and measurement error in the constructed unit-cost variable, and it is therefore attenuated relative to structural pass-through—[Subsection 5.3](#) already cautions that the regression is not an accounting decomposition. The counterfactual instead shocks an input price: raw poultry is a physical input requirement per pound of output, so a permanent increase in its price raises marginal cost one-for-one as an accounting matter, before any behavioral response. The two cost experiments in the paper are thus deliberately different: shocks to input prices load through this accounting identity, whereas shocks to product attributes, such as the distance scenario, load through the estimated cost surface, because an attribute has no market price of its own.

The plant-unavailability counterfactual removes products originating from a confidential set of southeastern counties chosen to represent weather-exposed poultry-processing capacity. Disclosure rules prevent me from reporting the exact county list used in the simulation. Instead, I describe the selection rule and report public examples of comparable counties in [Table A9](#). The confidential simulation counties were selected to resemble counties that contain poultry-processing capacity or related upstream poultry production and that have

historically experienced weather-related interruptions to poultry production, plant operations, road access, power service, or live-bird supply. The confidential set was fixed before solving the counterfactual equilibria and was not selected based on simulated welfare effects. I use these counties to define a stylized choice-set shock; the scenario is not an event-study design and does not require the disruption to coincide with a CFS reporting week. It is also an unconstrained spatial-reallocation benchmark: surviving plants expand at constant marginal cost, no capacity constraints bind, and excluded volume can flow to the residual outside option, so short-run capacity rigidities and contracting frictions would deepen the measured losses.

Large combinations are a recurring feature of this industry: between 1992 and 2007, 59 of 471 broiler-processing plants closed and 53 changed owners, and the 2006 acquisition of Gold Kist—then the third-largest producer—by second-ranked Pilgrim’s Pride shows that consolidation among leading firms is not hypothetical (Weng, Vukina, and Zheng, 2015). The merger scenario combines the third- and fourth-largest firms, where firm size is ranked by total CFS-expanded shipment value in the estimation sample. The merger is implemented by assigning the two firms’ products to a single owner in the ownership matrix, leaving demand and cost primitives unchanged. The scenario also holds product offerings fixed: no efficiencies, entry, or product repositioning offset the ownership change. Wollmann (2018) shows that product entry and exit can substantially moderate post-merger markup increases in the long run, so the price and welfare effects reported below are best read as short-run effects that abstract from offering adjustments.

6.3 Counterfactual Results

Figure 1 maps the state-level percentage change in total surplus for the four non-baseline counterfactuals. Darker shades indicate larger surplus losses. Two patterns are visible in both the maps and the state rows of Table A7. The raw-poultry cost shock is broad-based: nearly every destination state experiences a substantial total-surplus loss because the shock raises a core input cost for all suppliers. The plant-unavailability and merger scenarios are more spatially uneven. Their effects are largest where the unavailable origin-county products or merging firms account for important supply options, and smaller where buyers can substitute toward nearby non-affected plants. Corresponding state-level percentage-change maps for the FOB-plant unit value, cost, elasticity, inside sales, producer surplus, and downstream-buyer surplus appear in Figure A2, Figure A3, Figure A4, Figure A5, Figure A6, and Figure A7.

Table 3 reports five scenarios: the model-implied baseline, plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, increasing raw-poultry

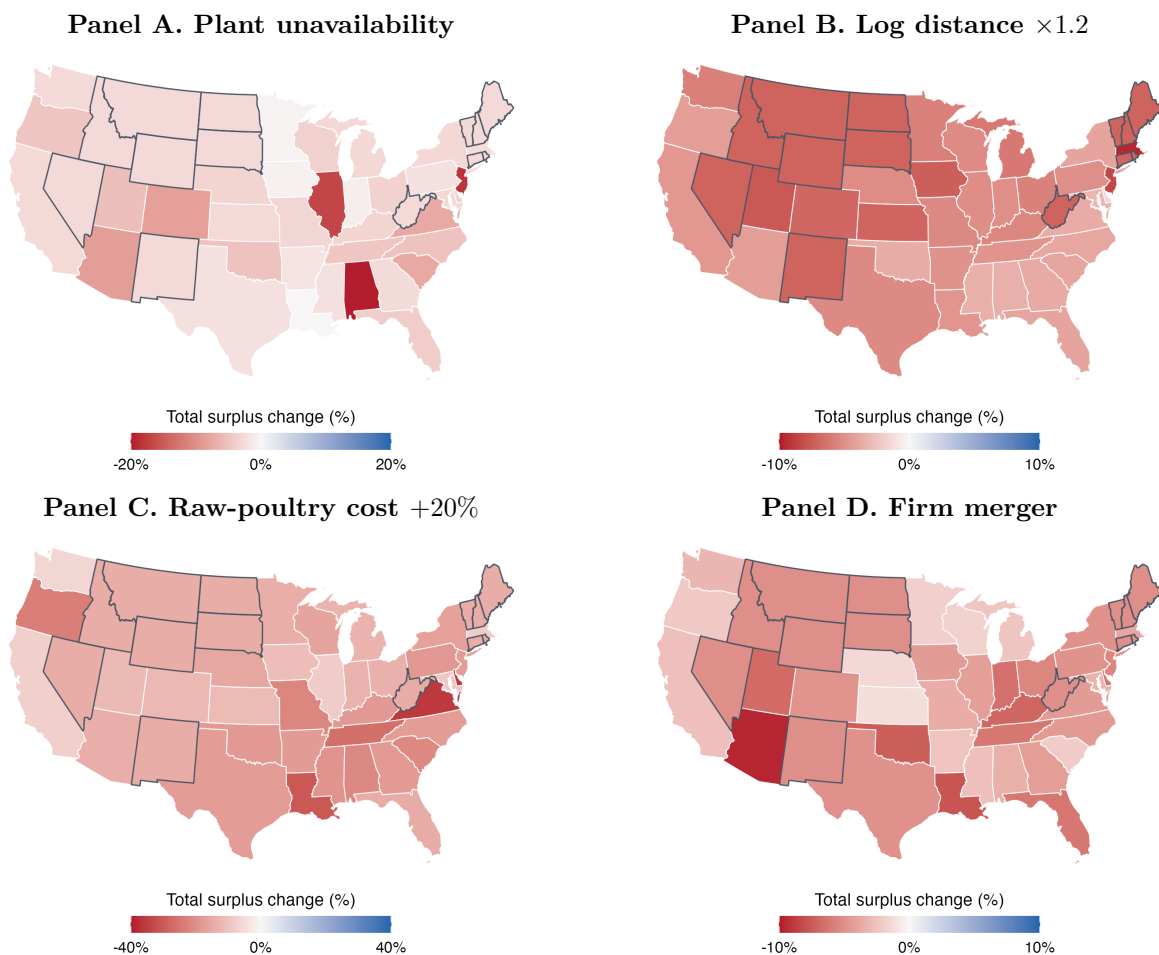


Figure 1: State-Level Total-Surplus Percentage Changes Relative to the Baseline
Notes: Total surplus equals producer surplus plus downstream-buyer surplus. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

cost by 20 percent, and merging the third- and fourth-largest firms. The first three columns report the average equilibrium FOB-plant unit value, recovered cost, and own-price elasticity; the remaining columns aggregate inside sales, producer surplus, downstream-buyer surplus, total surplus, and the change in total surplus relative to the baseline. The merger scenario changes ownership and pricing incentives but leaves cost primitives unchanged, whereas the distance and raw-poultry scenarios shift model-implied marginal costs and demand-relevant shipment attributes.

Table A7 reports the state-level results underlying these aggregate patterns; across scenarios, the ranking in Table 3 is sharp. The raw-poultry cost shock is by far the largest total-surplus shock, reducing total surplus by 15.90, or 17.72 percent. Plant unavailability and the log-distance shock are similar in aggregate magnitude, lowering total surplus by 4.32 and 4.15, respectively. The merger has the smallest total-surplus loss among the four non-baseline scenarios, 3.83, but it generates the sharpest redistribution between the two sides of the market because producer surplus rises while buyer surplus falls. The four shocks differ in primitive scale, so these comparisons should be read as a ranking of policy-relevant scenarios and a contrast of mechanisms, not as a size-normalized decomposition of spatial substitution.

Plant unavailability in the confidential weather-exposed county set primarily works through the product-choice set. The average FOB-plant unit value increases only slightly, from 1.443 to 1.448, and average recovered cost changes from 0.966 to 0.967. The small movement in average unit value and cost indicates that the aggregate welfare effect is not mainly a uniform marginal-cost increase. Instead, the scenario reduces the availability of nearby or otherwise attractive establishment–commodity options. Inside sales fall from 63.56 to 61.08, producer surplus falls from 34.39 to 33.04, and downstream-buyer surplus falls from 55.36 to 52.38. The total-surplus decline is therefore concentrated in lost trade and lost buyer surplus rather than in a large unit-value increase. In the state rows, the largest proportional losses occur in states such as Alabama, New Jersey, and Illinois, which is consistent with the map showing localized effects around destinations most exposed to the unavailable origin-county products.

The distance counterfactual is economically different because it raises effective shipment distance for all products by scaling log distance rather than by adding a fixed mileage increment. Average recovered cost increases from 0.966 to 0.990 and the average FOB-plant unit value increases from 1.443 to 1.469. Inside sales fall from 63.56 to 60.98, almost the same quantity contraction as in the plant-unavailability scenario. Producer surplus declines by 1.21 and buyer surplus declines by 2.95, yielding a total-surplus loss of 4.15, or 4.62 percent. The similarity between the plant-unavailability and distance-shock totals masks

different mechanisms: plant unavailability removes discrete supply options, whereas the distance shock makes the existing option set less attractive and more costly. The state-level table shows that percentage losses are more diffuse than in the plant-unavailability case, while the largest absolute losses appear in large baseline markets such as Texas, Arkansas, California, and Georgia.

The raw-poultry cost shock produces the most severe aggregate effect. A 20 percent increase in raw-poultry cost raises average recovered cost from 0.966 to 1.085, an increase of about 12 percent in the average marginal-cost measure, and raises the average FOB-plant unit value from 1.443 to 1.558, an increase of about 8 percent. The unit-value increase does not simply transfer surplus from buyers to producers. Instead, inside sales fall from 63.56 to 54.27, producer surplus falls from 34.39 to 27.28, and downstream-buyer surplus falls from 55.36 to 46.58. Both sides of the market lose because the input-cost shock compresses margins and contracts quantities. [Table A7](#) shows that the largest proportional total-surplus losses occur in states such as Delaware, Virginia, Louisiana, and Tennessee, while large poultry destination markets such as Virginia, Georgia, Arkansas, and Texas account for especially large absolute losses.

The merger scenario isolates the effect of ownership consolidation. Average recovered cost remains at 0.966 because the scenario does not alter technology or input prices. The average FOB-plant unit value rises from 1.443 to 1.542, inside sales fall from 63.56 to 57.70, and own-price elasticity becomes more negative in the new equilibrium. Producer surplus rises from 34.39 to 37.09, a gain of 2.70, because the merged firm internalizes substitution between the merging parties' products and optimally raises unit values. Downstream-buyer surplus falls from 55.36 to 48.83, a loss of 6.53, so total surplus still falls by 3.83. The merger therefore redistributes surplus from downstream buyers to producers while also generating deadweight loss: of the 6.53 buyer-surplus loss, 2.70 accrues to producers and 3.83 is lost. This decomposition is the clearest evidence in the table that ownership matters separately from costs: the merger increases the producer share of surplus even as it reduces the total size of the market. The state rows indicate that the proportional losses are largest in destinations where the merging firms appear to be particularly close or important competitors, including Arizona, Louisiana, Oklahoma, Kentucky, and Utah, while the largest absolute losses occur in larger destination markets such as Texas, Georgia, Florida, and Virginia.

Taken together, the table and maps show that the model distinguishes four adjustment channels. Plant unavailability reduces access to particular supply options, distance increases shipping frictions and model-implied marginal costs, raw-poultry input costs create a broad cost-push contraction, and ownership consolidation changes pricing incentives without changing marginal-cost primitives. Similar aggregate surplus losses can therefore come from dif-

ferent combinations of FOB-plant unit values, costs, quantities, and surplus redistribution.

7 Conclusion

The evidence shows how spatial shipment networks and downstream buyer heterogeneity shape competition and the incidence of supply disruptions in U.S. poultry procurement markets. The analysis builds a descriptive CFS-LBD poultry shipment network, uses a random-forest classifier to separate processor-to-retailer from processor-to-wholesaler links, and uses the resulting network measures to define the ToW demand shifter and motivate the LBD representative-buyer distribution. The structural analysis estimates preliminary logit benchmarks, a random-coefficients BLP demand model with buyer-type-specific price sensitivity, and a multi-product Bertrand supply side.

Product type, shipment distance, downstream channel composition, plant characteristics, and ownership all matter for procurement outcomes, while cold-chain requirements enter mainly through marginal cost rather than through demand. Demand is elastic on average, shipment distance raises costs for all products while lowering demand for differentiated ones, and the representative-buyer estimates indicate that retail and wholesale buyer types are more price sensitive than food-service buyers. The recovered supply side implies sizable markups, which makes ownership and substitution patterns central for welfare analysis.

Counterfactual simulations show that logistics shocks, raw-input cost shocks, plant unavailability, and ownership consolidation all reduce total surplus, but through different channels. The raw-poultry cost shock generates the largest aggregate loss and reduces both producer and downstream-buyer surplus. The merger scenario raises producer surplus while reducing buyer surplus and total surplus. Plant unavailability and distance shocks generate similar aggregate losses but operate through different mechanisms: plant unavailability removes discrete supply options, while distance shocks make the existing network less attractive and more costly.

The analysis also has deliberate limits. Because the work is conducted with restricted-use Census microdata inside a Federal Statistical Research Data Center (FSRDC), disclosure review constrains the number of reported robustness cells and alternative sample cuts. The baseline holds fixed the representative-buyer sampling rule, ownership grouping, and mapping from primitive shocks to costs or choice sets; the market-size normalization is varied directly in [Table A8](#). The counterfactuals are static partial-equilibrium simulations; they do not model plant entry, capacity adjustment, contract renegotiation, or long-run rewiring of procurement networks. These points are best read as limitations of the current evidence rather than as a menu of additional robustness tables.

More broadly, the paper illustrates how freight microdata can support empirical industrial organization. Linking shipment-network construction, machine-learning channel classification, LBD representative-buyer data, and BLP counterfactuals makes it possible to quantify how supply-chain geography and downstream buyer composition affect market power and the costs of supply disruptions. The same approach can be applied to other manufacturing and agricultural supply chains in which product flows, plant heterogeneity, and geographic competition jointly determine market outcomes.

Data Availability and Disclosure Statement

The empirical analysis relies on restricted-use Census microdata and linked establishment identifiers. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2521 and was reviewed for release under Disclosure Clearance Number CBDRB-FY25-0219. Public documentation for the CFS and Census business surveys is available from the U.S. Census Bureau and the Bureau of Transportation Statistics. The restricted microdata cannot be redistributed by the author. The replication package will contain code, synthetic or public-use examples where feasible, and disclosure-cleared tables and figures. All tables, figures, and other reported statistics in the public manuscript are subject to Census Bureau disclosure review and should be cited with the project and disclosure clearance numbers reported in this manuscript.

Required Census Bureau disclaimer. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2521. (CBDRB-FY25-0219)

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A Tables and Figures

A.1 Tables

Table A1: Full-Sample Fitted Confusion Matrix for Final PR–PW Classifier

Prediction	Reference class	
	To wholesaler	To retailer
To wholesaler	~1,200	~80
To retailer	~100	~15,500

Notes: Entries are rounded, disclosure-cleared fitted counts from the final random forest after refitting on the full mechanically labeled sample. This is not a held-out confusion matrix. Subsection 3.2 describes the 90–10 training-validation procedure used to select the specification and the subsequent full-sample refit. Accuracy on the 10 percent validation set used during specification selection was about 96 percent.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A2: Descriptive ZIP-Code Poultry Network Counts and Statistics by Link Type and Year

		Counts							Statistics					
Link	Year	Firms	Plants	Shipments	Src. ZIPs	Dest. ZIPs	Edges	Dist. wt.	Dist. val.	Density	Mean HHI	Wt. share	Val. share	
PR	2007	100	250	15,000	250	2,000	5,100	400	403	1.02%	0.266	88%	88%	
PR	2012	70	150	9,000	150	1,700	3,200	408	430	1.25%	0.256	91%	90%	
PR	2017	70	200	20,000	200	2,000	5,700	481	502	1.43%	0.203	87%	85%	
PW	2007	60	150	1,000	150	150	350	370	392	1.56%	0.740	6%	7%	
PW	2012	40	90	600	90	100	250	542	534	2.78%	0.680	5%	6%	
PW	2017	60	200	2,600	150	350	900	586	577	1.71%	0.495	10%	11%	
WR	2007	200	300	6,000	250	2,400	2,900	513	437	0.48%	0.133	6%	5%	
WR	2012	100	200	3,800	150	1,600	1,800	370	361	0.75%	0.197	4%	4%	
WR	2017	150	200	5,400	150	2,600	3,100	565	462	0.79%	0.315	3%	4%	

Notes: PR denotes processor-to-retailer shipments, PW denotes processor-to-wholesaler shipments, and WR denotes wholesaler-to-retailer or downstream-distribution shipments. Counts are rounded, disclosure-cleared descriptive network objects constructed before aggregation to the destination-county-year BLP estimation sample. Distances are miles and report shipment-distance averages weighted by shipment weight and shipment value. Network density is the number of observed directed edges divided by the product of source ZIPs and destination ZIPs. Mean HHI is computed across destination-ZIP shipment-weight shares for each source ZIP and then averaged using source-ZIP shipment weight. The final two columns report each link type's share of network shipment weight and value within the survey year.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A3: Descriptive Statistics for Estimation Variables

Variable	Mean	Std. dev.
FOB-plant unit value	1.4230	0.177
Share	0.0458	0.073
Log distance	6.0900	1.113
Frozen-shipment indicator	0.2769	0.451
Broiler indicator	0.6000	0.492
Processed-poultry indicator	0.2462	0.427
Private-carriage share (private)	0.1739	0.363
Wholesale-channel share (ToW)	0.0931	0.282
Log total production workers	7.9740	0.669
Log asset stock	10.7500	1.161
Materials unit cost (\$/lb)	0.6860	0.395
Supplemental unit cost (\$/lb)	0.1677	0.153
Labor unit cost (\$/lb)	0.1500	0.097
Capital expenditure rate	0.2434	0.242
Production-worker hourly pay	13.5000	3.136
Inventory-to-assets ratio	0.0309	0.040
Own-slaughter plant	0.6000	0.492
External-slaughter plant	0.1308	0.339
Rival product count	11.8500	10.521
Rival frozen-product count	4.7080	6.990
Rival broiler count	6.1320	6.023
Rival processed-product count	4.2940	7.861

Notes: The sample contains 6,500 product-market observations, 700 markets, and 400 products. Reported statistics are rounded in accordance with Census Bureau disclosure review, so distinct variables can display identical rounded values. The FOB-plant unit value and unit-cost variables are in constant 2017 dollars per pound; production-worker hourly pay is in constant 2017 dollars per hour.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A4: Sampled Shipments per Product-Market Cell

Shipments per cell	Share of cells	Mean shipments	Share of CFS-expanded weight
1–2	29.5%	1.5	4.5%
3–4	25.4%	3.6	12.4%
5–9	23.7%	6.7	26.2%
10–19	12.9%	13.0	24.6%
20 or more	8.5%	23.7	32.3%

Notes: The table reports the distribution of sampled CFS shipment records underlying the 6,500 product-market cells in the estimation sample. Across cells, the mean is about 6.6 shipments, the median is 4, the 90th and 95th percentiles are 15 and 22, and the maximum is about 55; the weight-weighted median cell is built from about 12 shipments. Shares and counts are rounded in accordance with Census Bureau disclosure review.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A5: BLP Mean Utility and Nested-Logit Parameters

Variable	Estimate
Constant	2.0820*** (0.6294)
Frozen-shipment indicator	0.3272 (0.2842)
Broiler product indicator	-0.9779*** (0.3638)
Processed-poultry indicator	-0.3449 (0.3992)
private	-0.1787 (0.2856)
ToW	1.1660** (0.4908)
Log shipment distance	-0.2103*** (0.0536)
Frozen $\times$ log distance	-0.0420 (0.0446)
Broiler $\times$ log distance	0.2164*** (0.0568)
Processed poultry $\times$ log distance	0.0919 (0.0628)
private $\times$ log distance	0.0345 (0.0500)
ToW $\times$ log distance	-0.1433* (0.0766)
2012 indicator	0.7888*** (0.0634)
2017 indicator	0.2600*** (0.0491)
Nested-logit parameter (ρ)	0.3229*** (0.0335)

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. The BLP nesting parameter ρ is reported with the demand-side parameters because it governs within-nest substitution rather than marginal cost. It uses the same broiler, processed-poultry, and other-poultry nests as the preliminary nested-logit model. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CDBRB-FY25-0219.

Table A6: Supply-Side Cost Parameters

Variable	Estimate	Variable	Estimate
Constant	0.6170*** (0.1060)	Log shipment distance	0.0280*** (0.0087)
Log total production workers	-0.0146** (0.0061)	Frozen $\times$ log distance	-0.0097 (0.0059)
Log asset stock	-0.0111*** (0.0042)	Broiler $\times$ log distance	-0.0027 (0.0079)
Materials unit cost (\$/lb)	0.0885*** (0.0083)	Processed poultry $\times$ log distance	0.0238*** (0.0090)
Supplemental unit cost (\$/lb)	0.1902*** (0.0202)	private $\times$ log distance	-0.0078 (0.0066)
Labor unit cost (\$/lb)	0.1893*** (0.0331)	ToW $\times$ log distance	0.0082 (0.0087)
Capital expenditure rate	0.0693*** (0.0167)	2012 indicator	0.0087 (0.0456)
Production-worker hourly pay	0.0003 (0.0013)	2017 indicator	0.2036*** (0.0385)
Inventory-to-assets ratio	0.5900*** (0.1002)	2012 indicator $\times$ log distance	0.0047 (0.0073)
Own-slaughter plant	-0.0122* (0.0071)	2017 indicator $\times$ log distance	-0.0239*** (0.0059)
External-slaughter plant	0.1076*** (0.0102)	Log destination-county population	0.0195*** (0.0034)
Frozen-shipment indicator	0.0821** (0.0375)	Rival product count	0.0015* (0.0009)
Broiler product indicator	-0.0509 (0.0494)	Rival frozen-product count	0.0024** (0.0010)
Processed-poultry indicator	-0.1453** (0.0567)	Rival broiler count	0.0023* (0.0013)
private	0.0442 (0.0377)	Rival processed-product count	-0.0026** (0.0011)
ToW	-0.0297 (0.0552)		

Notes: The estimation sample contains 6,500 product-market observations in 700 destination-county-year markets with 400 products. Standard errors clustered by destination county are in parentheses. The dependent variable is marginal cost in levels recovered from the supply-side first-order conditions; the cost parameters are estimated jointly with the demand parameters in the GMM system. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A7: State-Level Counterfactual Results by Scenario: Panel A. Baseline
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

State	FOB	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.426	0.9439	-3.063	6.465	2.806	1.567	2.525	4.092
Arizona	1.429	0.9961	-3.443	1.754	0.9064	0.4206	0.7731	1.194
Arkansas	1.416	0.9202	-3.377	21.25	5.808	3.504	4.893	8.397
California	1.436	0.9876	-3.199	12.45	5.809	2.922	5.119	8.041
Colorado	1.472	1.032	-3.33	1.099	0.4725	0.2199	0.4068	0.6267
Delaware	1.378	0.8349	-3.03	1.874	1.001	0.6699	0.917	1.587
Florida	1.444	0.9838	-3.388	7.924	3.838	1.977	3.357	5.334
Georgia	1.413	0.9164	-3.195	16.61	6.421	3.292	5.403	8.695
Illinois	1.5	1.042	-3.378	5.137	2.552	1.196	2.31	3.506
Indiana	1.436	0.9509	-3.073	0.8793	0.3923	0.2107	0.353	0.5637
Iowa	1.468	0.9756	-3.148	0.7941	0.4317	0.2497	0.4182	0.6679
Kansas	1.451	0.9696	-3.283	1.167	0.4553	0.2502	0.4062	0.6564
Kentucky	1.504	1.01	-3.088	1.144	0.6234	0.3201	0.554	0.8741
Louisiana	1.379	0.8809	-2.842	1.817	0.8343	0.492	0.7206	1.213
Maryland	1.418	0.9756	-3.244	2.362	1.348	0.6568	1.257	1.914
Massachusetts	1.495	1.079	-3.275	0.6518	0.3645	0.1646	0.3382	0.5028
Michigan	1.489	1.024	-3.194	2.72	1.246	0.6153	1.139	1.754
Minnesota	1.398	0.9422	-3.116	1.943	0.7739	0.3493	0.6316	0.9809
Mississippi	1.415	0.9023	-3.055	4.636	1.859	0.986	1.605	2.591
Missouri	1.421	0.917	-3.071	2.098	1.033	0.6326	0.9284	1.561
Nebraska	1.477	0.9665	-3.321	0.3374	0.1827	0.1218	0.1726	0.2944
New Jersey	1.429	0.9899	-3.081	1.673	0.7796	0.3709	0.6595	1.03
New York	1.422	0.9719	-3.142	6.686	3.026	1.388	2.564	3.952
North Carolina	1.408	0.9215	-2.99	4.412	2.267	1.202	1.999	3.201
Ohio	1.446	0.977	-3.061	2.608	1.336	0.6985	1.206	1.905
Oklahoma	1.45	0.9356	-3.065	0.6162	0.2857	0.171	0.2512	0.4222
Oregon	1.469	1.014	-2.753	0.8007	0.4284	0.2621	0.3756	0.6377
Pennsylvania	1.433	0.9741	-3.074	6.04	2.449	1.216	2.071	3.287
South Carolina	1.442	0.9649	-2.863	1.421	0.7128	0.3824	0.6252	1.008
Tennessee	1.451	0.9185	-2.932	2.246	1.077	0.6501	0.9306	1.581
Texas	1.397	0.9138	-3.318	14.8	6.018	3.225	5.115	8.34
Utah	1.506	1.062	-3.257	0.9695	0.5153	0.2391	0.4481	0.6872
Virginia	1.418	0.9137	-2.535	10.43	3.52	2.711	3.099	5.81
Washington	1.478	1.02	-3.432	1.122	0.6144	0.2815	0.5487	0.8302
Wisconsin	1.478	0.9795	-2.966	1.548	0.8063	0.4583	0.7324	1.191
Rest of America	1.472	0.9683	-3.287	1.169	0.5678	0.3156	0.5066	0.8222

Table A7: State-Level Counterfactual Results by Scenario: Panel B. Plant unavailability
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

State	FOB	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.412	0.9364	-2.985	6.465	2.471	1.274	2.008	3.282
Arizona	1.429	0.9923	-3.368	1.754	0.85	0.3966	0.6936	1.09
Arkansas	1.419	0.9181	-3.365	21.25	5.668	3.455	4.78	8.235
California	1.443	0.9903	-3.183	12.45	5.63	2.863	4.949	7.812
Colorado	1.493	1.045	-3.325	1.099	0.427	0.202	0.3718	0.5738
Delaware	1.398	0.8573	-2.901	1.874	0.9796	0.6492	0.8797	1.529
Florida	1.453	0.9885	-3.375	7.924	3.697	1.926	3.192	5.118
Georgia	1.417	0.9156	-3.164	16.61	6.253	3.25	5.214	8.464
Illinois	1.511	1.058	-3.572	5.137	2.248	0.9421	1.971	2.913
Indiana	1.442	0.9542	-3.075	0.8793	0.3874	0.2087	0.3499	0.5586
Iowa	1.476	0.9871	-3.107	0.7941	0.4304	0.2476	0.4164	0.664
Kansas	1.457	0.9708	-3.268	1.167	0.4413	0.2469	0.3907	0.6376
Kentucky	1.501	1.011	-3.047	1.144	0.6068	0.3091	0.5356	0.8447
Louisiana	1.372	0.8733	-2.839	1.817	0.8328	0.4915	0.7188	1.21
Maryland	1.427	0.9795	-3.191	2.362	1.317	0.6458	1.21	1.856
Massachusetts	1.48	1.061	-3.201	0.6518	0.3586	0.1629	0.3271	0.49
Michigan	1.486	1.02	-3.178	2.72	1.217	0.5974	1.105	1.702
Minnesota	1.397	0.9403	-3.107	1.943	0.7705	0.3491	0.6271	0.9762
Mississippi	1.408	0.8914	-3.017	4.636	1.815	0.9809	1.552	2.533
Missouri	1.424	0.9117	-3.073	2.098	1.002	0.6216	0.8903	1.512
Nebraska	1.477	0.94	-3.324	0.3374	0.1733	0.1192	0.1651	0.2843
New Jersey	1.434	0.9956	-3.323	1.673	0.6832	0.3035	0.5377	0.8412
New York	1.427	0.9748	-3.127	6.686	2.944	1.363	2.47	3.833
North Carolina	1.424	0.9299	-2.956	4.412	2.156	1.173	1.867	3.04
Ohio	1.449	0.9773	-3.006	2.608	1.28	0.6956	1.141	1.837
Oklahoma	1.456	0.937	-3.001	0.6162	0.2719	0.1673	0.2332	0.4005
Oregon	1.492	1.025	-2.7	0.8007	0.4073	0.2582	0.3487	0.6069
Pennsylvania	1.438	0.9776	-3.043	6.04	2.395	1.201	2.017	3.218
South Carolina	1.469	0.9798	-2.844	1.421	0.6638	0.3629	0.568	0.9309
Tennessee	1.429	0.8916	-2.835	2.246	1.029	0.6412	0.8643	1.506
Texas	1.392	0.9102	-3.295	14.8	5.904	3.18	4.973	8.153
Utah	1.507	1.062	-3.223	0.9695	0.4932	0.2304	0.4192	0.6496
Virginia	1.422	0.9235	-2.597	10.43	3.357	2.487	2.885	5.372
Washington	1.487	1.028	-3.412	1.122	0.5984	0.2776	0.5289	0.8065
Wisconsin	1.488	0.978	-2.946	1.548	0.7789	0.453	0.6945	1.147
Rest of America	1.494	0.9668	-3.288	1.169	0.5467	0.3112	0.4882	0.7994

Table A7: State-Level Counterfactual Results by Scenario: Panel C. Log distance $\times 1.2$
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

State	FOB	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.449	0.966	-3.086	6.465	2.723	1.521	2.428	3.949
Arizona	1.467	1.03	-3.459	1.754	0.87	0.4063	0.7355	1.142
Arkansas	1.439	0.9431	-3.398	21.25	5.55	3.349	4.636	7.985
California	1.469	1.017	-3.214	12.45	5.541	2.851	4.816	7.667
Colorado	1.507	1.064	-3.358	1.099	0.444	0.2077	0.3751	0.5828
Delaware	1.395	0.8505	-3.049	1.874	0.9859	0.663	0.8989	1.562
Florida	1.474	1.011	-3.411	7.924	3.688	1.921	3.198	5.119
Georgia	1.436	0.9384	-3.217	16.61	6.225	3.191	5.18	8.371
Illinois	1.528	1.066	-3.397	5.137	2.436	1.153	2.172	3.325
Indiana	1.464	0.9756	-3.099	0.8793	0.3738	0.2035	0.3314	0.5349
Iowa	1.49	0.997	-3.194	0.7941	0.4094	0.2347	0.3841	0.6188
Kansas	1.483	0.9996	-3.33	1.167	0.4271	0.2358	0.3739	0.6097
Kentucky	1.53	1.037	-3.119	1.144	0.5963	0.3061	0.5198	0.8259
Louisiana	1.403	0.9029	-2.881	1.817	0.8028	0.471	0.6835	1.155
Maryland	1.446	1	-3.281	2.362	1.309	0.6428	1.209	1.852
Massachusetts	1.517	1.096	-3.247	0.6518	0.3377	0.154	0.3004	0.4544
Michigan	1.52	1.051	-3.24	2.72	1.188	0.5861	1.06	1.646
Minnesota	1.415	0.958	-3.157	1.943	0.7355	0.334	0.59	0.924
Mississippi	1.432	0.9195	-3.073	4.636	1.802	0.9579	1.542	2.5
Missouri	1.448	0.9419	-3.097	2.098	0.9886	0.6041	0.8731	1.477
Nebraska	1.503	0.9885	-3.337	0.3374	0.1752	0.1171	0.1619	0.279
New Jersey	1.442	1.001	-3.083	1.673	0.7275	0.3454	0.5983	0.9437
New York	1.45	0.9966	-3.171	6.686	2.922	1.345	2.448	3.793
North Carolina	1.428	0.9405	-3.007	4.412	2.198	1.168	1.908	3.076
Ohio	1.472	1.001	-3.106	2.608	1.275	0.6682	1.127	1.795
Oklahoma	1.477	0.9607	-3.066	0.6162	0.2757	0.1667	0.2401	0.4068
Oregon	1.499	1.041	-2.789	0.8007	0.4112	0.2552	0.355	0.6102
Pennsylvania	1.458	0.9967	-3.085	6.04	2.346	1.165	1.956	3.121
South Carolina	1.461	0.9832	-2.892	1.421	0.6914	0.3688	0.5985	0.9673
Tennessee	1.479	0.9468	-2.952	2.246	1.037	0.6214	0.8828	1.504
Texas	1.427	0.9414	-3.349	14.8	5.704	3.109	4.789	7.898
Utah	1.541	1.092	-3.264	0.9695	0.4814	0.2271	0.4076	0.6347
Virginia	1.443	0.9371	-2.541	10.43	3.414	2.621	2.973	5.594
Washington	1.509	1.045	-3.476	1.122	0.5822	0.272	0.5099	0.7819
Wisconsin	1.5	1.001	-2.986	1.548	0.7706	0.4388	0.6892	1.128
Rest of America	1.49	0.9857	-3.289	1.169	0.5352	0.3004	0.463	0.7634

Table A7: State-Level Counterfactual Results by Scenario: Panel D. Raw-poultry cost +20%
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

State	FOB	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.566	1.088	-3.522	6.465	2.312	1.184	2.029	3.213
Arizona	1.524	1.087	-3.795	1.754	0.7978	0.3545	0.6696	1.024
Arkansas	1.556	1.054	-3.623	21.25	4.916	2.804	4.086	6.89
California	1.507	1.058	-3.353	12.45	5.353	2.61	4.804	7.414
Colorado	1.546	1.106	-3.668	1.099	0.4226	0.184	0.3656	0.5496
Delaware	1.654	1.132	-3.773	1.874	0.7095	0.41	0.5964	1.006
Florida	1.548	1.091	-3.68	7.924	3.403	1.606	2.947	4.553
Georgia	1.566	1.071	-3.564	16.61	5.402	2.638	4.487	7.125
Illinois	1.521	1.063	-3.6	5.137	2.403	1.09	2.126	3.216
Indiana	1.556	1.065	-3.414	0.8793	0.3431	0.1802	0.308	0.4882
Iowa	1.553	1.081	-3.417	0.7941	0.3994	0.2031	0.3905	0.5936
Kansas	1.594	1.101	-3.507	1.167	0.3948	0.2218	0.3577	0.5795
Kentucky	1.562	1.077	-3.508	1.144	0.5285	0.2641	0.4498	0.7139
Louisiana	1.563	1.063	-3.522	1.817	0.6208	0.3367	0.5062	0.8429
Maryland	1.533	1.086	-3.416	2.362	1.218	0.6028	1.129	1.732
Massachusetts	1.507	1.094	-3.583	0.6518	0.3435	0.1478	0.3133	0.4611
Michigan	1.547	1.091	-3.562	2.72	1.123	0.5088	1.012	1.521
Minnesota	1.548	1.086	-3.468	1.943	0.6665	0.301	0.5404	0.8414
Mississippi	1.604	1.092	-3.545	4.636	1.51	0.7989	1.292	2.091
Missouri	1.577	1.082	-3.456	2.098	0.8444	0.4905	0.7282	1.219
Nebraska	1.563	1.076	-3.605	0.3374	0.1616	0.1001	0.1484	0.2485
New Jersey	1.544	1.099	-3.576	1.673	0.6618	0.2909	0.5575	0.8484
New York	1.544	1.091	-3.511	6.686	2.562	1.138	2.141	3.279
North Carolina	1.551	1.075	-3.408	4.412	1.923	0.9464	1.689	2.635
Ohio	1.535	1.073	-3.42	2.608	1.188	0.5906	1.055	1.646
Oklahoma	1.542	1.05	-3.447	0.6162	0.2551	0.1329	0.2117	0.3446
Oregon	1.555	1.102	-3.111	0.8007	0.3539	0.1894	0.2985	0.4879
Pennsylvania	1.555	1.091	-3.464	6.04	2.059	0.9839	1.696	2.68
South Carolina	1.597	1.12	-3.379	1.421	0.5878	0.289	0.4997	0.7887
Tennessee	1.59	1.086	-3.557	2.246	0.8522	0.4507	0.7105	1.161
Texas	1.567	1.083	-3.648	14.8	5.075	2.543	4.312	6.855
Utah	1.538	1.1	-3.544	0.9695	0.4663	0.2122	0.3937	0.6059
Virginia	1.574	1.077	-3.43	10.43	2.615	1.582	2.138	3.72
Washington	1.572	1.112	-3.65	1.122	0.5764	0.2617	0.5149	0.7766
Wisconsin	1.556	1.076	-3.38	1.548	0.7198	0.3659	0.6324	0.9983
Rest of America	1.567	1.075	-3.446	1.169	0.506	0.2636	0.4406	0.7042

Table A7: State-Level Counterfactual Results by Scenario: Panel E. Firm merger
Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

State	FOB	Cost	Elas.	Mkt.	Inside	PS	BS	TS
Alabama	1.512	0.9439	-3.14	6.465	2.591	1.686	2.264	3.95
Arizona	1.583	0.9961	-3.799	1.754	0.7304	0.4908	0.5875	1.078
Arkansas	1.452	0.9202	-3.463	21.25	5.492	3.642	4.549	8.191
California	1.506	0.9876	-3.28	12.45	5.437	3.106	4.718	7.824
Colorado	1.546	1.032	-3.505	1.099	0.426	0.244	0.3526	0.5966
Delaware	1.567	0.8349	-3.248	1.874	0.8619	0.7342	0.7497	1.484
Florida	1.568	0.9838	-3.602	7.924	3.338	2.201	2.8	5.001
Georgia	1.493	0.9164	-3.308	16.61	5.828	3.579	4.733	8.312
Illinois	1.569	1.042	-3.525	5.137	2.312	1.334	2.021	3.355
Indiana	1.568	0.9509	-3.28	0.8793	0.3408	0.2349	0.2922	0.5271
Iowa	1.58	0.9756	-3.265	0.7941	0.39	0.2721	0.3659	0.638
Kansas	1.49	0.9696	-3.326	1.167	0.4408	0.255	0.3935	0.6485
Kentucky	1.675	1.01	-3.283	1.144	0.5301	0.3599	0.4525	0.8124
Louisiana	1.497	0.8809	-3.097	1.817	0.6995	0.5466	0.571	1.118
Maryland	1.514	0.9756	-3.355	2.362	1.249	0.7153	1.135	1.85
Massachusetts	1.628	1.079	-3.432	0.6518	0.3323	0.1833	0.2998	0.4831
Michigan	1.538	1.024	-3.283	2.72	1.178	0.6536	1.058	1.712
Minnesota	1.419	0.9422	-3.19	1.943	0.7433	0.3669	0.5952	0.9621
Mississippi	1.482	0.9023	-3.142	4.636	1.746	1.036	1.485	2.521
Missouri	1.495	0.917	-3.165	2.098	0.9487	0.6725	0.8318	1.504
Nebraska	1.569	0.9665	-3.364	0.3374	0.1764	0.1241	0.1659	0.29
New Jersey	1.5	0.9899	-3.304	1.673	0.6935	0.4093	0.5643	0.9736
New York	1.548	0.9719	-3.3	6.686	2.719	1.529	2.227	3.756
North Carolina	1.523	0.9215	-3.108	4.412	2.044	1.304	1.752	3.056
Ohio	1.579	0.977	-3.227	2.608	1.185	0.7707	1.029	1.8
Oklahoma	1.625	0.9356	-3.302	0.6162	0.2427	0.1883	0.2026	0.3909
Oregon	1.544	1.014	-2.811	0.8007	0.4045	0.2723	0.3512	0.6235
Pennsylvania	1.516	0.9741	-3.219	6.04	2.202	1.332	1.794	3.126
South Carolina	1.521	0.9649	-2.898	1.421	0.6787	0.3969	0.589	0.9859
Tennessee	1.615	0.9185	-3.067	2.246	0.9323	0.7075	0.7747	1.482
Texas	1.492	0.9138	-3.484	14.8	5.342	3.486	4.44	7.926
Utah	1.637	1.062	-3.556	0.9695	0.4438	0.2723	0.3681	0.6404
Virginia	1.514	0.9137	-2.545	10.43	3.173	2.868	2.683	5.551
Washington	1.56	1.02	-3.561	1.122	0.5699	0.3021	0.502	0.8041
Wisconsin	1.53	0.9795	-3.02	1.548	0.7706	0.4744	0.6945	1.169
Rest of America	1.571	0.9683	-3.486	1.169	0.503	0.3447	0.4358	0.7805

Table A8: Sensitivity to the Market-Size Normalization

κ	Mkt.	Inside share	Elas.	Margin (%)	PS	BS	TS	ΔTS (%) by scenario			
								Plant	Distance	Raw cost	Merger
0.6	176.93	0.359	-3.180	32.8	34.12	54.74	88.86	-4.57	-4.54	-17.13	-4.02
0.7 (baseline)	151.65	0.419	-3.135	33.1	34.39	55.36	89.75	-4.81	-4.62	-17.72	-4.27
0.8	132.69	0.479	-3.041	33.7	34.73	58.24	92.97	-5.02	-4.71	-18.17	-4.41

Notes: Each row re-estimates the full model and re-solves all four counterfactual scenarios with the outside-option normalization in Appendix B.2 set to κ in place of the baseline 0.7. Mkt. is aggregate market size in billion lb and Inside share is aggregate inside sales divided by market size; both follow mechanically from the normalization. Elas. is the average own-price elasticity; Margin is the average unit-value–cost difference as a percent of the FOB-plant unit value. PS, BS, and TS are baseline producer, downstream-buyer, and total surplus in billions of dollars under each normalization. The ΔTS columns report each scenario’s total-surplus change relative to that row’s baseline, in percent. Scenario rankings and signs are unchanged across normalizations.

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A9: Public Examples of Weather-Disrupted Poultry-Processing and Production Counties

County	FIPS	Poultry-processing anchor	Weather or disruption motivation
Blount County, AL	01009	Tyson Blountsville chicken-processing plant.	April 2011 tornado outbreak: Tyson reported that the Blountsville plant was idled after losing power.
Marshall County, AL	01095	Tyson Albertville chicken-processing plant and Wayne Farms Albertville processing facilities.	April 2011 tornado outbreak: Tyson reported that Albertville was idled after losing power; Wayne Farms reported Albertville-area processing facilities idled for seven days.
Morgan County, AL	01103	Wayne Farms Decatur processing facilities.	April 2011 tornado outbreak: Wayne Farms reported that Decatur-area processing facilities were idled for seven days by widespread power outages.
Tuscaloosa County, AL	01125	Peco Foods Tuscaloosa poultry facility.	April 2011 tornado outbreak: Meat+Poultry reported that the Tuscaloosa facility was heavily damaged.
Forsyth County, GA	13117	Tyson Cumming chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Cumming chicken plant in response to the storm.
Terrell County, GA	13273	Tyson Dawson chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Dawson chicken plant in response to the storm.
Dooly County, GA	13093	Tyson Vienna chicken plant.	Hurricane Irma in September 2017: Tyson shut down the Vienna chicken plant in response to the storm.
Lenoir County, NC	37107	Sanderson Farms Kinston poultry-processing facility.	Hurricane Matthew in October 2016: Reuters reported severe North Carolina poultry losses, including about 250,000 Sanderson broilers lost in the state.
Lexington County, SC	45063	House of Raeford West Columbia poultry plant and South Carolina grower region.	Hurricane Matthew in October 2016: House of Raeford reported partial contract-farm flock losses and weather-related road-access problems.
Jasper County, MS	28061	Peco Foods Bay Springs poultry-processing and slaughter facility.	April 2011 tornado outbreak: the National Weather Service reported significant poultry-farm damage along the Smith–Jasper–Clarke tornado track.

Notes: The table reports public examples of southeastern counties that combine poultry-processing capacity or upstream poultry production with documented weather-related disruption. It does not disclose the confidential county set used in the plant-unavailability simulation. The examples illustrate the type of geography used to motivate the stylized choice-set shock: counties where storms, flooding, power outages, road-access problems, or live-bird losses plausibly interrupted poultry production or processing. Sources: [Tyson Foods \(2011\)](#), [MEAT+POULTRY \(2011\)](#), [Powder & Bulk Solids \(2017\)](#), [Ag-Canada/Reuters \(2016\)](#), [WATTagNet \(2016\)](#), [National Weather Service Jackson \(2011\)](#), [FSIS \(n.d.a\)](#), [FSIS \(n.d.b\)](#), and [FSIS \(n.d.c\)](#).

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Table A10: Main Variables Used in Estimation

Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

Variable	Definition
FOB-plant unit value	CFS-sampling-weight-expanded shipment value divided by expanded shipment weight, aggregated to the product-market-year level and converted to constant 2017 dollars per pound. CFS shipment value excludes freight charges and excise taxes.
Share	Product-market quantity divided by adjusted market size.
Log distance	Log of shipment distance weighted by CFS-expanded shipment weight.
Frozen	Indicator equal to one for SCTG 05122 and zero for SCTG 05121.
Broiler	Indicator for plants assigned to the broiler product category using manufacturing product-line information.
Processed poultry	Indicator for plants assigned to the processed-poultry product category; the omitted category is other poultry plants.
private	Private-carriage share: the share of shipments transported using shipper-owned trucks.
ToW	Product-market CFS-expanded weight share of processor-origin shipments classified as processor-to-wholesaler (PW) using the PR-PW classifier.
type_R	Sampled-record indicator equal to one for retail representative buyer records; food service is the omitted buyer type.
type_W	Sampled-record indicator equal to one for wholesale representative buyer records; food service is the omitted buyer type.
sampling_weight	LBD representative-buyer sampling size equal to establishment employment times the buyer-type throughput ratio.
tpw	Total production workers.
tae	Plant asset stock measure.
wp	Production-worker payroll per production-worker hour.
te	Capital expenditures divided by plant assets.
mc	Materials unit cost, equal to materials cost divided by CFS-expanded shipment weight and reported in \$/lb.
sc	Supplemental input unit cost, equal to supplemental input cost divided by CFS-expanded shipment weight and reported in \$/lb.
wc	Labor unit cost, equal to labor cost divided by CFS-expanded shipment weight and reported in \$/lb.
tie	Total inventories divided by ending total assets.
Packaging input intensity	CMF materials-detail packaging cost divided by CFS-expanded shipment weight (\$/lb); enters the excluded demand instruments.
Raw-poultry input intensity	CMF materials-detail raw-poultry cost divided by CFS-expanded shipment weight (\$/lb); enters the excluded demand instruments and is the cost term raised by 20 percent in the raw-poultry counterfactual.
Poultry-parts input intensity	CMF materials-detail processed-poultry cost divided by CFS-expanded shipment weight (\$/lb); enters the excluded demand instruments.
TYPOP104	Indicator for poultry or meat products processed chiefly from animals slaughtered in the same establishment.
TYPOP106	Indicator for poultry or meat products processed chiefly from animals slaughtered elsewhere.
Turkey indicator	Indicator equal to one for plants whose CMF product line is turkey; used only in constructing the local differentiation instruments (turkey plants belong to the other-poultry category).
Log destination-county population	Log of population in the market's destination county and year; a supply-side control in the marginal-cost equation.
n_rival	Number of competing products in the same destination-county-year market.
n_rival_frozen	Number of competing frozen products in the same market.
n_rival_broiler	Number of competing broiler products in the same market.
n_rival_processed	Number of competing processed-poultry products in the same market.

A.2 Figures

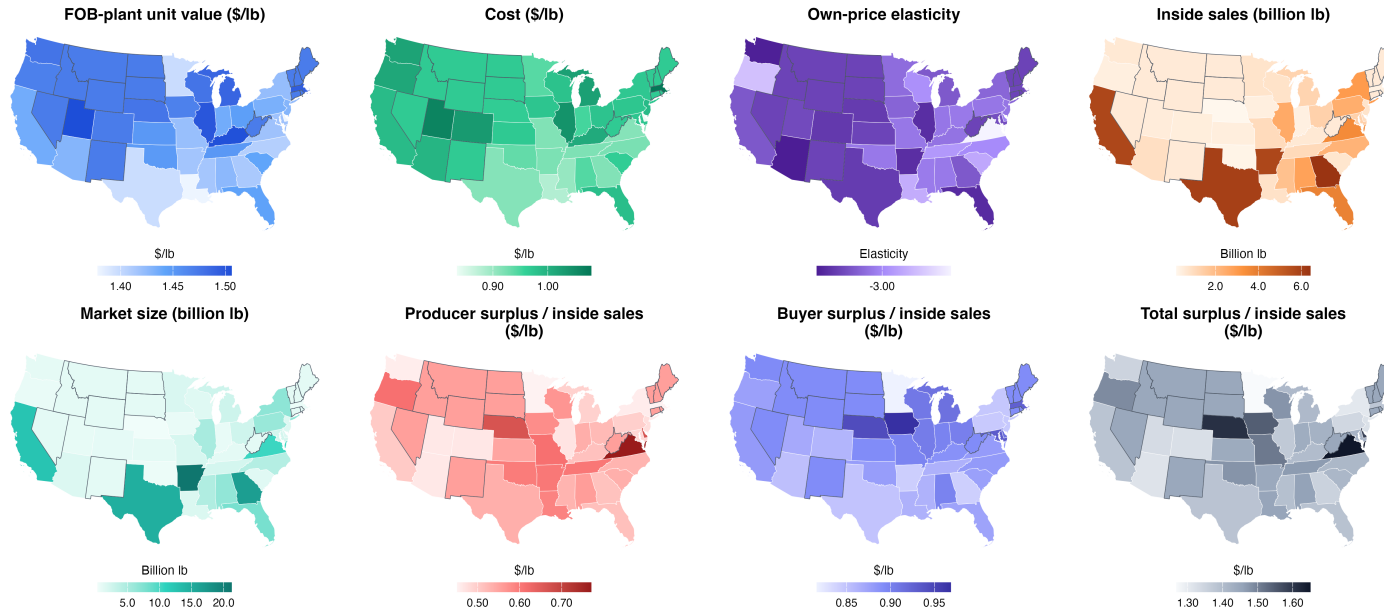


Figure A1: Model-Implied Baseline State-Level Distributions

Notes: Panels map the model-implied baseline levels of the FOB-plant unit value, recovered cost, own-price elasticity, inside sales, market size, and the three surplus measures normalized by inside sales. The FOB-plant unit value, recovered cost, and the three surplus-per-inside-sales measures are reported in \$/lb; inside sales and market size are reported in billion lb. The total-surplus-per-pound panel equals total surplus divided by inside sales, where total surplus is producer surplus plus downstream-buyer surplus. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

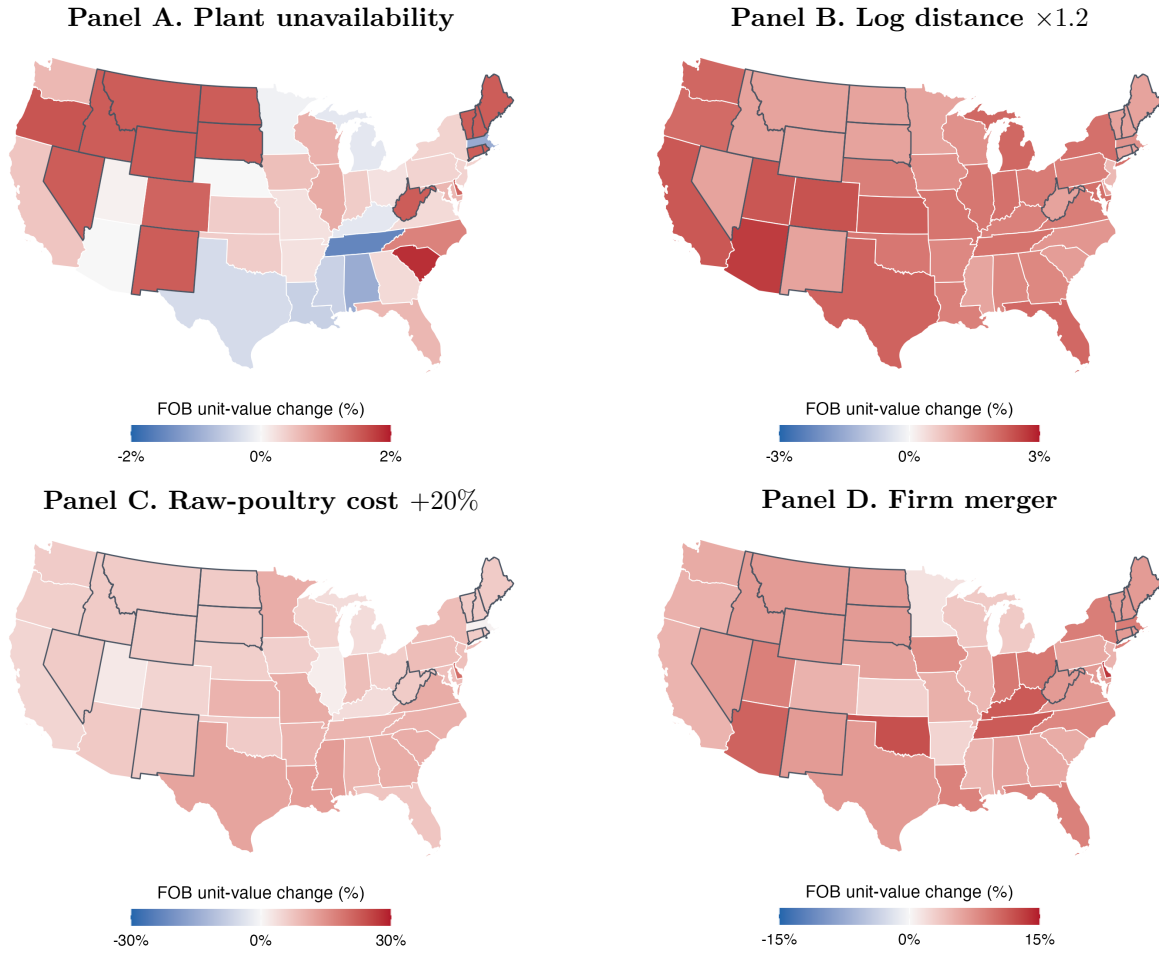


Figure A2: State-Level FOB-Plant Unit-Value Percentage Changes Relative to the Baseline
Notes: The FOB-plant unit value is the average equilibrium unit value in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

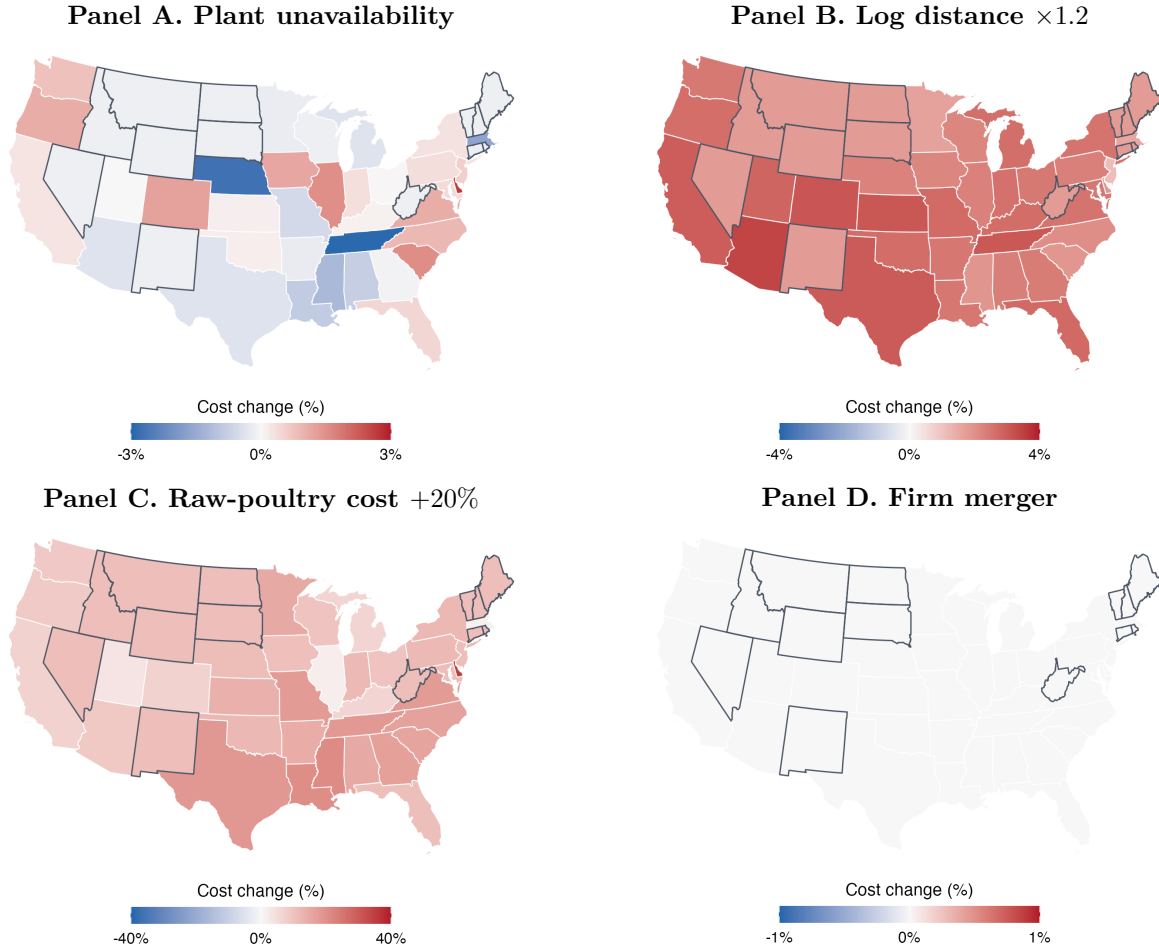


Figure A3: State-Level Cost Percentage Changes Relative to the Baseline

Notes: Cost is the recovered average marginal-cost measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

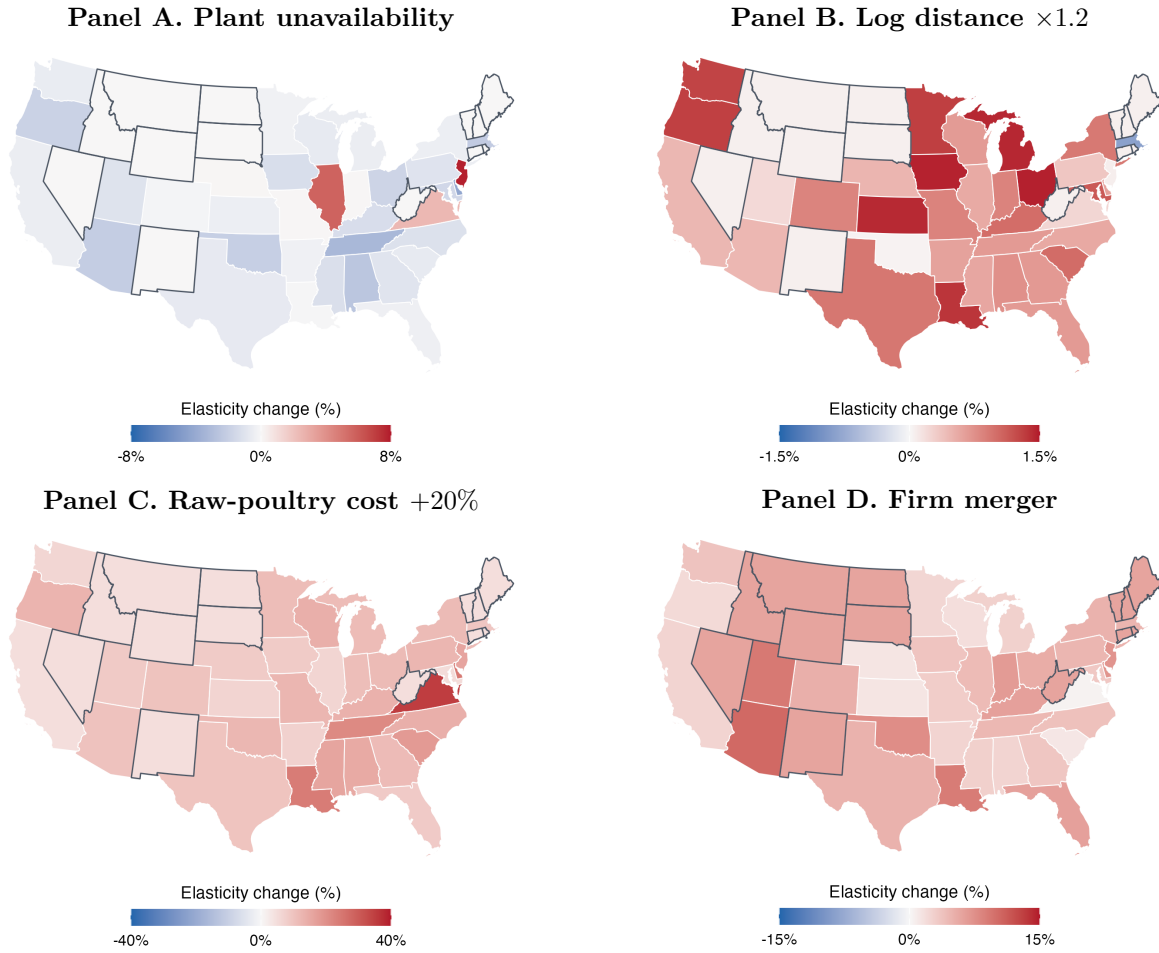


Figure A4: State-Level Own-Price Elasticity Percentage Changes Relative to the Baseline
Notes: Elasticity is the average own-price elasticity in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

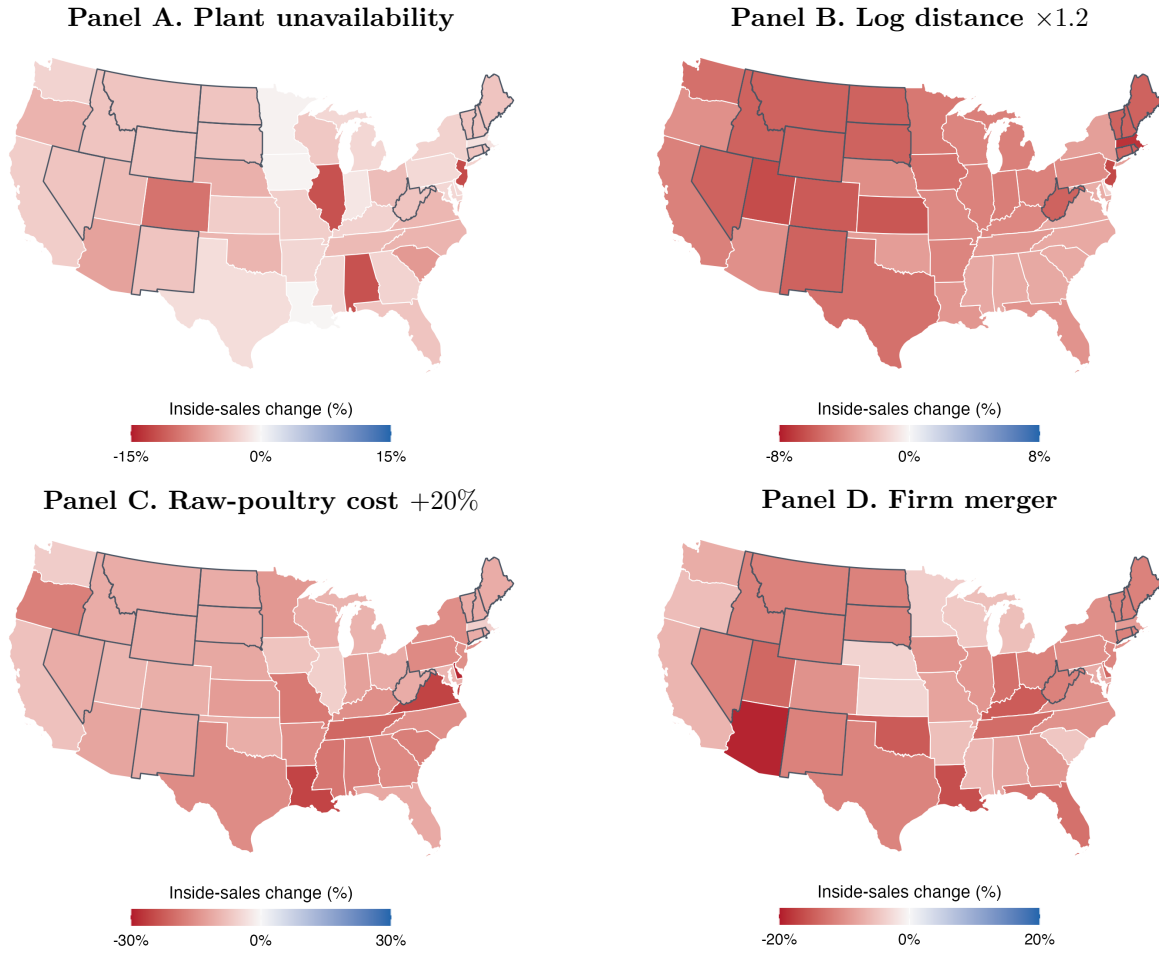


Figure A5: State-Level Inside-Sales Percentage Changes Relative to the Baseline

Notes: Inside sales aggregate the equilibrium inside quantity measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

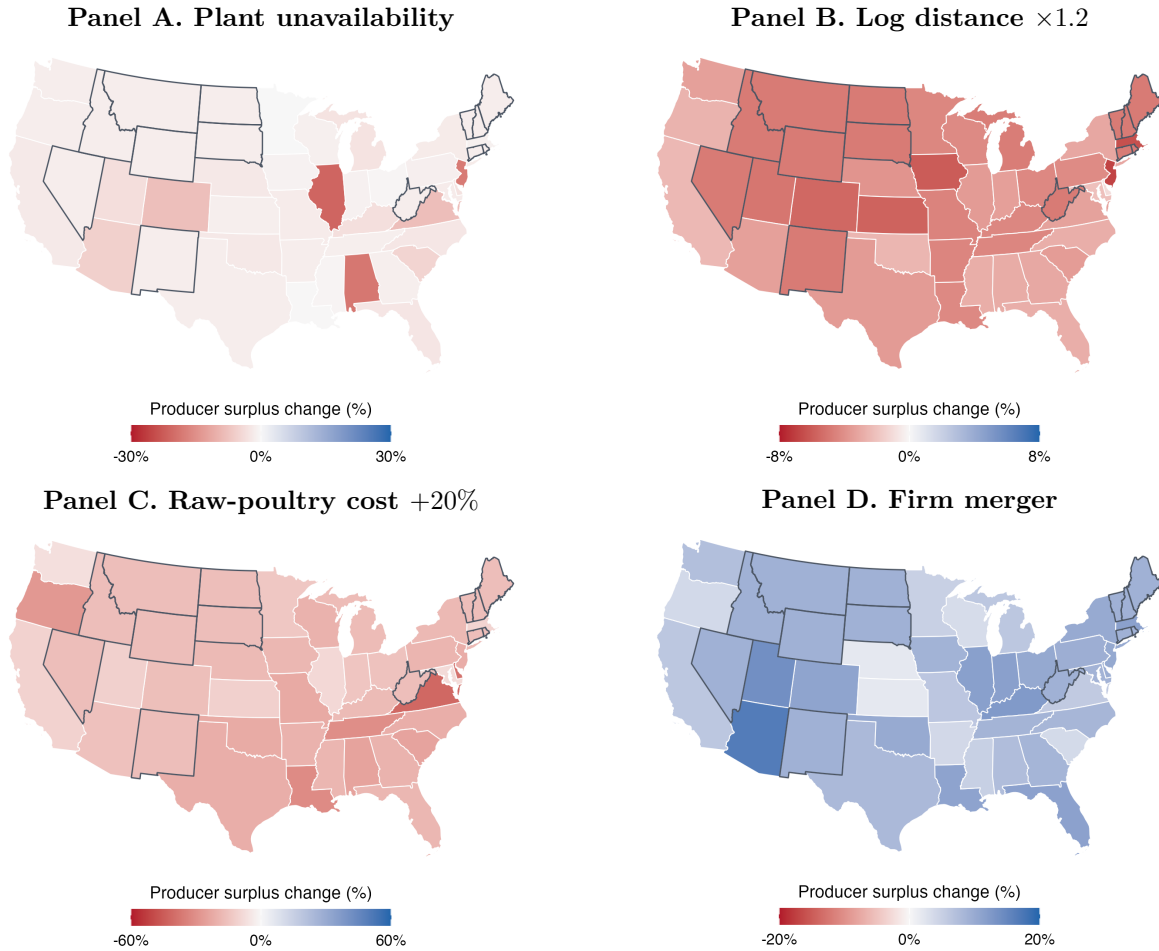


Figure A6: State-Level Producer-Surplus Percentage Changes Relative to the Baseline

Notes: Producer surplus is the model-implied producer surplus in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

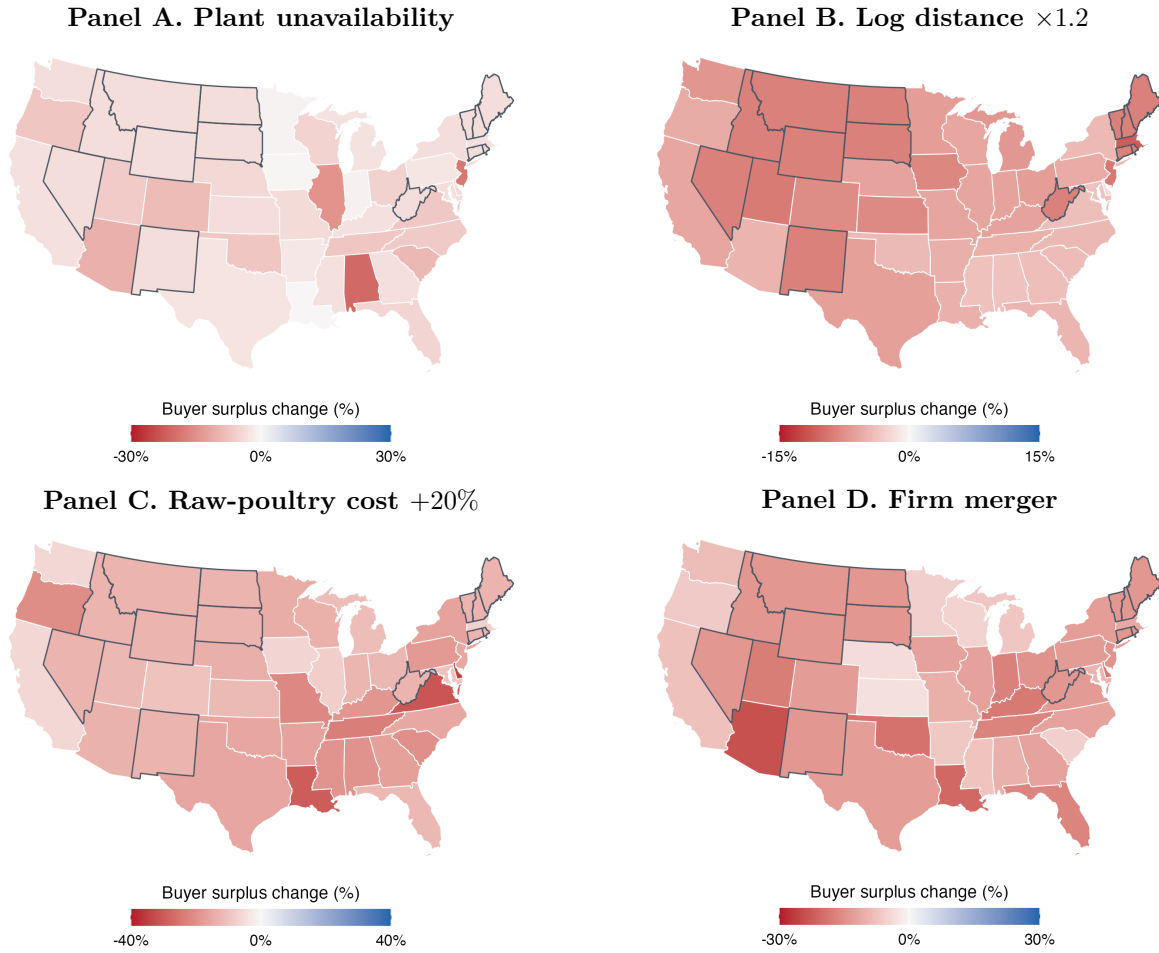


Figure A7: State-Level Downstream-Buyer-Surplus Percentage Changes Relative to the Baseline

Notes: Downstream-buyer surplus is the BLP log-sum surplus measure in the state row. Panels A–D show percentage changes relative to the model-implied baseline under plant unavailability in the confidential weather-exposed county set, scaling log distance by 1.2, a 20 percent increase in raw-poultry cost, and a merger of the third-largest and fourth-largest firms, respectively. Maps display the contiguous United States. States without separate state rows are colored with the Rest of America aggregate value and outlined in dark gray. Disclosure: FSRDC Project Number 2521; Disclosure Clearance Number CBDRB-FY25-0219.

B Data Construction Details

This appendix reports implementation details for the descriptive shipment network, market-size proxy, and LBD representative-buyer sample. These objects support the variables defined in Section 3, but the structural model uses the product-market variables and representative buyer records described in the main text.

B.1 Descriptive Poultry Shipment Network

This appendix records the construction of the descriptive ZIP-code poultry shipment network summarized in Subsection 3.3, using the channel assignments produced in Subsection 3.2. Unlike the destination-county-year markets maintained in the structural model, the descriptive network keeps origin and destination ZIP codes so that shipment links can be characterized before aggregation.

The network uses the same poultry SCTG codes as the structural sample: 05121 for fresh or chilled poultry and 05122 for frozen poultry. I begin with poultry shipment records originating from meat processors and grocery-related wholesalers. The descriptive extract contains roughly 63,000 poultry shipments across the 2007, 2012, and 2017 CFS waves. In a typical wave, about 200 processors and 200 wholesalers appear as CFS shippers. Because the CFS is a shipment survey rather than a census of all poultry establishments, the network objects should be interpreted as disclosure-cleared descriptive summaries of observed shipment links rather than as the universe of all poultry flows.

Shipments are organized into three directed link types. Processor-origin shipments use the PR and PW classification described in Subsection 3.2. Wholesaler-origin shipments are classified as wholesaler-to-retailer or downstream-distribution links, denoted WR. After assigning shipment channels, I aggregate records to directed origin-ZIP–destination-ZIP edges by survey year and link type. Let $L \in \{PR, PW, WR\}$ index the link type and let t index the survey year. Table A2 reports the number of firms, plants, shipment records, source ZIP codes, destination ZIP codes, unique directed edges, weighted distances, density, HHI, and shipment shares for each (L, t) cell. For link type L in year t , network density is computed as

$$\text{Density}_{Lt} = \frac{E_{Lt}}{S_{Lt}D_{Lt}},$$

where E_{Lt} is the number of observed directed ZIP-to-ZIP edges, S_{Lt} is the number of source ZIPs, and D_{Lt} is the number of destination ZIPs. The HHI statistic is first computed within

each source ZIP using destination-ZIP shipment-weight shares,

$$\text{HHI}_{zLt} = \sum_d \left(\frac{\tilde{w}_{zdLt}}{\sum_{d'} \tilde{w}_{zd'Lt}} \right)^2,$$

where $\tilde{w}_{zdLt}$ is the CFS-expanded shipment weight on directed edge $z \rightarrow d$. The HHI is then averaged across source ZIPs using source-ZIP CFS-expanded shipment weight. The final two columns of [Table A2](#) report each link type’s share of total network shipment weight and value in the survey year.

The descriptive network shows that PR links dominate observed poultry flows in all three waves, accounting for roughly 87–91 percent of shipment weight and 85–90 percent of shipment value. PW links become more important over time: their shipment-weight share rises from about 6 percent in 2007 to about 10 percent in 2017, and their value share rises from about 7 percent to 11 percent. WR links move in the opposite direction, falling to about 3–4 percent of shipment weight and value by 2017. Distance patterns also differ by link type. Processor-origin links become longer on average, especially PW links, while WR distances are more volatile. These patterns motivate the model’s use of shipment distance, downstream channel exposure, and local buyer composition, but the descriptive ZIP network is not itself the maintained BLP market definition.

B.2 Market-Size Construction

The final market-size measure in Subsection 3.5 combines CFS inside shipments, population-based poultry availability, and downstream retail and wholesale activity from ARTS/AWTS. The ARTS/AWTS variables are scale proxies for potential downstream poultry demand; they do not replace the CFS quantities used to form product-level inside shares. I first construct a population-based scale,

$$M_m^{\text{pop}} = \text{pop}_m \times \overline{\text{consumption}}_{t(m)},$$

where $\overline{\text{consumption}}_{t(m)}$ is national annual per-capita poultry availability in the CFS survey year of market m , measured in pounds and constructed from public chicken and turkey availability series ([National Chicken Council, n.d.](#)).

Because market m is a destination-county-year cell, let $c(m)$ denote its destination county and let $t(m)$ denote its year. I use county c and year y to describe the auxiliary ARTS/AWTS estimates before mapping them back to market m . For each ARTS/AWTS establishment record i , let w_i denote the survey weight WGT, and let z_i be one of total sales CSAL, total inventories CTINV, or operating expenses COPEX. Records are assigned to counties using re-

ported FIPS codes or a ZIP-to-county crosswalk and are retained for the meat and poultry retail/wholesale NAICS cells used in the downstream calibration. The direct county-year survey estimate for variable z is

$$\hat{Z}_{cy}^{\text{dir}}(z) = \sum_{i \in \mathcal{S}_{cy}} w_i z_i, \quad z \in \{\text{CSAL}, \text{CTINV}, \text{COPEX}\}.$$

Annual ARTS/AWTS samples are sparse at the county level, so I stabilize the direct estimates using a five-year window around the year of market m , $\mathcal{W}(t(m)) = \{t(m) - 2, t(m) - 1, t(m), t(m) + 1, t(m) + 2\}$. The market year receives weight 3, adjacent years receive weight 2, and the two outer years receive weight 1. Let a_y denote this time weight and let $r_{c(m)y}$ be a reliability weight based on the number of usable survey observations in county $c(m)$ and year y . The pooled direct estimate assigned to market m is

$$\bar{Z}_m^{\text{dir}}(z) = \frac{\sum_{y \in \mathcal{W}(t(m))} a_y r_{c(m)y} \hat{Z}_{c(m)y}^{\text{dir}}(z)}{\sum_{y \in \mathcal{W}(t(m))} a_y r_{c(m)y}}.$$

For counties with little or no direct survey support, I use the LBD county-year universe to construct small-area predictions. The auxiliary LBD variables are total employment, total payroll, and the number of downstream establishments in the same county-year. On county-years with usable ARTS/AWTS direct estimates, I fit log per-worker outcomes of the form

$$\log \left(\frac{\hat{Z}_{cy}^{\text{dir}}(z) + 1}{\text{EMP}_{cy}^{\text{LBD}} + 1} \right) = X'_{cy} \beta_z + \mu_c + \eta_y + \varepsilon_{cy},$$

where X_{cy} contains LBD employment, payroll-per-worker, and establishment-count controls, and μ_c and η_y are county and year effects. The fitted values generate model-based county-year predictions that I map back to each market as $\bar{Z}_m^{\text{model}}(z) = \bar{Z}_{c(m),t(m)}^{\text{model}}(z)$. I combine direct and model-based information with an empirical shrinkage weight,

$$\bar{Z}_m(z) = \lambda_m \bar{Z}_m^{\text{dir}}(z) + (1 - \lambda_m) \bar{Z}_m^{\text{model}}(z), \quad \lambda_m = \frac{n_m^{\text{win}}}{n_m^{\text{win}} + 10},$$

where n_m^{win} is the number of usable survey observations in the five-year window. Finally, I benchmark the county estimates so that their state-year, or national-year when state support is insufficient, totals match the corresponding pooled ARTS/AWTS survey totals. For state $s(m)$,

$$Z_m^{\text{bench}}(z) = \bar{Z}_m(z) \frac{\sum_{c \in s(m)} \bar{Z}_{c,t(m)}^{\text{dir}}(z)}{\sum_{c \in s(m)} \bar{Z}_{c,t(m)}(z)}.$$

The variables in the market-size regression are then

$$\begin{aligned}\text{meat_sales}_m &= Z_m^{\text{bench}}(\text{CSAL}), \\ \text{meat_inventories}_m &= Z_m^{\text{bench}}(\text{CTINV}), \\ \text{operating_expenses}_m &= Z_m^{\text{bench}}(\text{COPEX}).\end{aligned}$$

The adjusted market-size proxy is obtained by regressing observed inside shipments on the population and downstream-scale variables,

$$\begin{aligned}R_m &= \zeta_0 + \zeta_1 M_m^{\text{pop}} + \zeta_2 \text{meat_sales}_m + \zeta_3 \text{meat_inventories}_m \\ &\quad + \zeta_4 \text{operating_expenses}_m + \nu_m,\end{aligned}\tag{33}$$

and denoting the fitted value by $\widehat{M}_m^{\text{adj}}$. The final baseline market-size proxy is

$$M_m = \frac{\max\{\widehat{M}_m^{\text{adj}}, R_m\}}{0.7}.\tag{34}$$

The max operator ensures that market size is not smaller than observed CFS-expanded inside shipments R_m . The denominator is a baseline normalization that leaves room for an outside option, since the retained CFS scope is not a complete census of poultry procurement. I use this market-size proxy to form product shares and the outside share in Subsection 3.5.

B.3 LBD Buyer-Agent Sampling

The random-coefficients model integrates over a market-specific sample of downstream buyers. In this paper, a representative buyer record is a numerical integration node drawn from LBD establishments in a destination-county-year market. It is not an individual household consumer and it is not an additional raw data source.

Let a index LBD establishments located in market m . I first classify each eligible establishment into one of three mutually exclusive buyer types using harmonized NAICS codes in the LBD extract:

$$T_{am} = \begin{cases} W, & \text{if NAICS is 424410, 424420, 424470, or 424440,} \\ R, & \text{if NAICS is 445110, 455211, or 445240,} \\ F, & \text{if NAICS begins with 7225.} \end{cases}\tag{35}$$

Here F denotes food-service establishments, R denotes retailers, and W denotes wholesalers. Food service is the omitted buyer type in the demand specification.

The LBD contains many small establishments in each county-year. Treating each establishment as equally representative would overstate the role of small restaurants and convenience stores relative to large retailers and wholesale distributors. I therefore use a probability-proportional-to-size (PPS) sampling weight that combines establishment employment with a buyer-type-specific measure of poultry throughput per employee. The throughput ratios are normalized so that food service equals one:

$$\tau_F = 1, \quad \tau_R = 2.89, \quad \tau_W = 38.12. \quad (36)$$

These ratios are calibration inputs for sampling representative buyer records; they are not estimated demand parameters. They enter the model only through the sampling probabilities in (41) below, so they shape the market-specific buyer-type distribution rather than the demand coefficients directly; Appendix E lists this calibration among the constructed inputs on which the reported standard errors condition. The numerical ratios are constructed from public national throughput-per-worker benchmarks. For buyer type $g \in \{F, R, W\}$, let $\tilde{\tau}_g = Q_g/E_g$, where Q_g is annual poultry pounds handled and E_g is employment in the corresponding public NAICS industries. Food-service and retail pounds use 2017 chicken and turkey per-capita availability, together with ERS away-from-home shares of 47 percent for chicken and 30 percent for turkey (U.S. Department of Agriculture, Economic Research Service, 2021; National Chicken Council, n.d.):

$$Q_F = N^{\text{US}}(92.1 \times 0.47 + 16.4 \times 0.30) \approx 15.67 \text{ billion lb}, \quad (37)$$

$$Q_R = N^{\text{US}}(92.1 \times 0.53 + 16.4 \times 0.70) \approx 19.60 \text{ billion lb}, \quad (38)$$

where $N^{\text{US}} = 325$ million approximates the 2017 U.S. population. Dividing by 2017 Economic Census employment gives $\tilde{\tau}_F = 15.67 \text{ billion lb}/10,717,088 \approx 1,462 \text{ lb per employee}$ for food service (NAICS 7225) and $\tilde{\tau}_R = 19.60 \text{ billion lb}/4,630,685 \approx 4,232 \text{ lb per employee}$ for the retail NAICS set used in the public calibration. For wholesale, I sum Economic Census poultry product-line sales across the relevant wholesale NAICS industries, 24.759 billion dollars, and convert dollars to pounds using \$1.498 per pound, a composite of the 2017 broiler wholesale and retail price composites:

$$Q_W = \frac{24.759 \text{ billion dollars}}{1.498 \text{ dollars/lb}} \approx 16.53 \text{ billion lb}. \quad (39)$$

With wholesale employment $E_W = 296,581$, this gives $\tilde{\tau}_W = 16.53 \text{ billion lb}/296,581 \approx 55,728 \text{ lb per employee}$ (U.S. Census Bureau, 2024c). Normalizing by the food-service benchmark yields $\tau_R = 4,232/1,462 \approx 2.89$ and $\tau_W = 55,728/1,462 \approx 38.12$.

For establishment a in market m , the sampling size is

$$A_{am} = \text{emp}_{am} \times \tau_{T_{am}}, \quad (40)$$

where emp_{am} is LBD employment and $\tau_{T_{am}}$ is the relative throughput per employee for the establishment's buyer type. This weighting scheme makes a wholesale employee represent substantially more potential poultry flow than a retail or food-service employee, while still allowing local employment to determine how much mass each establishment receives within its type.

For each market m , I draw 1,000 representative buyer records with replacement from the eligible establishment set $\mathcal{A}_m = \{a : T_{am} \in \{F, R, W\}\}$. The probability that establishment a is selected on any draw is

$$\Pr(i = a \mid m) = \frac{A_{am}}{\sum_{b \in \mathcal{A}_m} A_{bm}}. \quad (41)$$

The draw is therefore a probability-proportional-to-size approximation to the local buyer distribution. After sampling, the 1,000 representative buyer records are used as the numerical integration nodes in the BLP demand system. Because the sampling probabilities already incorporate employment and throughput differences, the integration weights are equal within a market:

$$\psi_{im} = \frac{1}{1,000}, \quad i = 1, \dots, 1,000. \quad (42)$$

The 700 estimation markets therefore generate 700,000 representative buyer records.

B.4 Instrument Set

The demand system treats the FOB-plant unit value as endogenous. The excluded demand instruments used in Section 4 are built from three groups:

1. Plant cost and scale shifters: materials, supplemental-input, and labor unit costs; production-worker payroll per hour; production employment; plant asset stock; capital expenditure relative to assets; inventory relative to assets; input-intensity measures for packaging, raw poultry, and poultry parts, defined as the corresponding CMF materials-detail costs per pound of CFS-expanded shipment weight; and indicators for own-slaughter and external-slaughter processing plants.
2. Market-structure shifters: counts of rival products, rival frozen products, rival broiler products, rival processed-poultry products, and distinct firms in the destination-county-year market.

3. Local differentiation instruments: Gandhi–Houde local differentiation measures constructed from frozen status, shipment distance, broiler status, turkey status, and processed-poultry status (Gandhi and Houde, 2019).

The product’s own log shipment distance is an included demand characteristic and therefore serves as its own instrument. The distance-based local differentiation measures are separate excluded instruments for the endogenous FOB-plant unit value. Turkey status is an indicator equal to one for plants whose CMF product line is turkey; because turkey plants belong to the other-poultry category in the demand and supply specifications, the indicator enters only the construction of the differentiation instruments. The supply-side moments use the exogenous cost shifters, product and plant-type indicators, shipment characteristics, year indicators, destination population, and rival-count variables in the marginal-cost specification.

C Variable Definitions

Table A10 reports the main variables used in the demand, supply, and preliminary regression specifications. The table is intended as a compact dictionary for variables whose construction is described in Section 3 and Appendix B. It orders variables by their economic role rather than by source file: shipment-based variables, representative-buyer variables, plant characteristics, and market-structure instruments.

Several variables are defined at the product-market level. A product is an establishment–commodity pair defined by the establishment and its SCTG code, and a market is a destination-county-year cell. Product-market FOB-plant unit values, quantities, shares, shipment distance, private-carriage shares, and the wholesale-channel share ToW are all constructed from CFS records after multiplying shipment-level values and physical shipment weights by the CFS sampling weight. The product-level indicators **Frozen**, **Broiler**, and **Processed poultry** combine SCTG information with CMF product-line information, so temperature status and plant product specialization enter separately.

The representative-buyer variables are defined at the sampled-record level within each market. The indicators **type_R** and **type_W** identify sampled retail and wholesale representative buyer records, with food service as the omitted group. The sampling size combines LBD employment with the buyer-type throughput-per-employee calibration described in Appendix B.3; it affects which establishments are represented in the numerical integration sample but is not itself an estimated demand parameter.

The supply-side controls are plant-level characteristics merged from Census manufacturing records. The unit-cost variables **mc**, **sc**, and **wc** are reported in constant 2017 dollars per pound and correspond to materials, supplemental inputs, and labor, respectively; the

mnemonic `mc` refers to materials cost and is distinct from the marginal cost mc_{jm} in the supply model. The rival-count variables summarize the composition of the choice set in each destination-county-year market and enter the preliminary and structural specifications as market-structure controls or instruments.

D State-Level Counterfactual Results

Figure A1 first displays the model-implied baseline state-level distributions of the FOB-plant unit value and cost (both in \$/lb), own-price elasticity, inside sales and market size (both in billion lb), and producer, downstream-buyer, and total surplus normalized by inside sales (in \$/lb). Table A7 reports the state-level rows underlying Figure 1 and the aggregate counterfactual summary in Table 3. The table reports destination states by name after converting the original destination-state FIPS codes; Rest of America corresponds to the original FIPS 99 aggregate. The table is organized by scenario so that each block can be compared directly with the model-implied baseline. Within a state, the relevant welfare comparison is the change in the TS (total-surplus) column relative to the baseline row for the same state; the main text summarizes these state-level changes in percentage and absolute terms.

In Table A7, FOB denotes the FOB-plant unit value and is reported in \$/lb; cost is also reported in \$/lb; Mkt. and Inside denote market size and inside sales in billion lb; and PS, BS, and TS denote producer surplus, downstream-buyer surplus, and total surplus in billions of dollars. Within each state row, FOB, cost, and elasticity are averages across product-market cells weighted by CFS-expanded quantity. The plant-unavailability scenario makes products from the confidential weather-exposed county set unavailable; the distance scenario scales log distance by 1.2; the cost scenario raises raw-poultry cost by 20 percent; and the firm-merger scenario combines the third-largest and fourth-largest firms, ranked by total CFS-expanded shipment value in the estimation sample.

D.1 Public Examples for the Plant-Unavailability Scenario

Table A9 reports public examples of southeastern counties with poultry-processing capacity or upstream poultry production that experienced weather-related interruptions. The table is not the confidential county list used to run the plant-unavailability simulation. The examples clarify the type of county used to motivate the weather-exposed choice-set shock; the counterfactual itself remains a stylized simulation rather than an event-study estimate.

E Validation Under Disclosure Constraints

All analyses are conducted with restricted-use Census microdata inside a Federal Statistical Research Data Center. The validation strategy is therefore deliberately narrow: I report only checks that are essential for credibility and can be summarized in disclosure-cleared aggregate form. I do not present a broad grid of alternative specifications that would require many additional sample cells, state-by-industry cuts, or unpublished establishment-level diagnostics. The central question is not whether every plausible variant has been exhausted, but whether the two constructed ingredients that carry the empirical design—the PR–PW channel classifier and the unit-value-endogenous demand specification—have enough support to interpret the estimates.

The first check is PR–PW classifier assessment. Subsection 3.2 describes a 90–10 training-validation partition: candidate specifications are compared by ten-fold cross-validation within the training set, the validation set is used to select the hyperparameter specification, and the final random forest is then refit on the full labeled sample. The disclosure-cleared confusion matrix reports fitted classifications from this final refit and is not a held-out matrix. Accuracy on the 10 percent validation set was about 96 percent, with comparable values for the other validation-set metrics. This check is necessary because ToW and the descriptive shipment network depend on the classifier; in particular, it asks whether the classifier can recover the smaller processor-to-wholesaler class rather than simply predict the dominant processor-to-retailer channel. The high fitted accuracy and meaningful recovery of the PW class support treating the random-forest assignment as a data-construction device rather than as a separately estimated economic mechanism, subject to the transferability assumption in Subsection 3.2.

The second check is demand-identification validation. The preliminary OLS, IV, and nested-logit estimates in Table 1 are used as diagnostics for the structural model: the IV coefficient on the FOB-plant unit value is more negative than OLS, the nested-logit parameter supports stronger within-nest substitution, and the excluded instruments enter through the same unit-value-endogeneity channel used in the BLP specification. The released evidence on instrument relevance is the OLS-to-IV movement together with cluster-robust weak-instrument statistics: the Montiel Olea–Pflueger effective first-stage F is 77.62 for the IV logit, well above its critical value, and the Kleibergen–Paap rk Wald F of 58.19 for the nested-logit benchmark, reported as a descriptive diagnostic. These diagnostics are more important than a large set of alternative counterfactuals because rejected demand identification would undermine every surplus calculation. Additional sensitivity analyses, such as alternative representative-buyer sampling rules, ownership groupings, and shock magnitudes, are

better treated as limitations or future extensions unless disclosure-cleared aggregate versions are needed during review.

One additional limitation follows from the same disclosure-constrained design. Several inputs are constructed before structural estimation, including the wholesale-channel share ToW , the market-size proxy M_m , the throughput calibration used in representative-buyer sampling, and the PPS representative-buyer draws. Of these, the market-size normalization is examined directly: [Table A8](#) re-estimates the model under outside-option normalizations of 0.6 and 0.8 and shows that scenario signs, rankings, and magnitudes are essentially unchanged. [Table A4](#) similarly documents the sampling depth behind the product-market cells; thin cells carry little CFS-expanded weight. The reported standard errors condition on these constructed inputs rather than propagating all first-stage prediction, calibration, and sampling uncertainty. For the same reason, the counterfactual outcomes in [Section 6](#) are reported as point values evaluated at the estimated parameters: simulation-based uncertainty measures would require many additional disclosure cells, and the scenario comparisons should be read as model-implied magnitudes rather than as estimates with released sampling intervals.